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Review article

Exploring wind farm reliability: Key concepts, databases and fault Trees

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ABSTRACT

This review delves into the reliability of wind farms, aiming to identify and assess the reliability of critical components which are essential for their dependable operation. By synthesizing data from various surveys, this study isolates key components crucial to the sustained functionality of wind farms. Each identified component undergoes a thorough assessment using fault tree analysis, providing a detailed evaluation of its impact on the overall reliability of wind energy systems. To achieve this goal, this study explores systematically the failure rates and downtimes of crucial elements within wind farms, such as wind turbines, transformers, HVDC/HVAC systems, and circuit breakers. For an in-depth exploration of transformer failures, numerous surveys from Japan, Brazil, the United Kingdom, and CIGRE have been reviewed. The examination extends to cable systems, encompassing AC and DC configurations in marine and submarine environments, incorporating findings from 59 utility entities across 19 countries. Furthermore, the study scrutinizes circuit breaker failure rates by analyzing data from 27 utilities representing 13 countries. Additionally, to assess the reliability of wind farms, this review introduces a Fault Tree Analysis, categorizing wind farms subsystems and providing insights for reliability allocation and operation and maintenance management. The findings not only shed light on strategies to enhance wind farm reliability but also offer a valuable road map for future research directions and industry best practices.

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1. Introduction

The reliability of a composite power system revolves around evaluating the capacity of both the generation and transmission systems to deliver sufficient electrical energy to the main load centers [7]. This approach has demonstrated efficacy across different sectors, encompassing generation source planning, transmission infrastructure development, transmission operation planning, and operational reserve assessment [8,9]. In recent times, there has been a global surge in the rapid and extensive development of wind power generation as a means to address challenges related to environmental pollution and the sustainable supply of energy [10]. Because of the unpredictable and fluctuating nature of wind speed, the output power from wind farms is stochastic and significantly differs from that of conventional power units [11]. Consequently, there is an urgent requirement to investigate the reliability assessment of composite power systems incorporating the integration of wind power.

The success of a wind energy project hinges on its reliability. Insufficient reliability may lead to frequent breakdowns and demanding extensive maintenance, while high reliability can mitigate breakdown costs and frequency, the associated expenses may be excessively prohibitive. Breakdowns and maintenance constitute substantial portions of a Wind farm's operation and maintenance expenditure. Furthermore, the reliability of wind farms has a direct impact on the overall system performance and power output, resulting in additional costs due to lost revenue [12,13]. Data on the reliability of a wind farm include the historical records of failures, repairs, and downtimes for the wind farm subassemblies. A comprehensive comprehension of wind farm reliability plays a pivotal role in crafting effective operation and maintenance (O&M) strategies, ultimately leading to enhanced wind farm performance.

In recent years, offshore wind energy has undergone substantial expansion, with an annual cumulative installed capacity projected to continue growing. Since 2011, the annual cumulative capacity has surged by more than fourfold, reaching 18.8 GW in 2017. Concurrently, advancements in turbine technologies have facilitated the deployment of larger turbines in more distant offshore locations. However, the offshore wind industry is currently grappling with a range of operational challenges. These challenges include the imperative to enhance component reliability and the necessity to implement and develop operation and maintenance strategies aimed at reducing overall costs [14].

The wind energy industry has employed fault tree analysis as a robust methodology for assessing reliability and safety. This approach is chosen for its practicality in identifying critical components and discerning the root causes of failures within the complex context of wind energy systems [15–17].

This review introduces a FTA of a wind farm, categorizing the system into several major subsystems based on their functions. These subsystems are then dissected and analyzed through the development of subtrees. The outcomes of this analysis offer insights for reliability allocation as well as operation and maintenance (O&M) management. However, it is noted that obtaining historical failure data for basic events is a prerequisite for implementing quantitative FTA. Yet, due to the limited number of wind farm samples, acquiring such data proves challenging.

This review contributes significantly in several key aspects:

- The document presents an extensive examination of current reliability data sources, outlining the commonalities and distinctions among various populations of wind turbines. It assesses the failure rates and downtimes of individual wind turbine components, utilizing a unified taxonomy derived from multiple data sources. The goal is to pinpoint critical subassemblies by evaluating their failure rates and downtimes.
- Also an in-depth investigation into the number of failures and identification of critical components in transformers has been conducted, drawing insights from surveys conducted in diverse global regions. The examination encompasses data from Japan, Brazil, and the United Kingdom, providing a multifaceted perspective on transformer performance. Additionally, insights from CIGRE transformer reliability database have been incorporated, enriching the analysis with international standards and practices. This amalgamation of survey findings from different geographical contexts and industry perspectives sheds light on the nuanced aspects of transformer failures and critical components in a global context.
- It also delves into the extensive evaluation of cable systems, encompassing both AC and DC configurations, across marine and submarine environments and varying voltage levels. Drawing data from a broad spectrum, the study incorporates findings from 59 utility entities spanning 19 diverse countries, providing a

Abbreviations	Symbols
HVDC - High Voltage Direct Current	λ - Failure rate per year
HVAC - High Voltage Alternating Current	T_i - Overall duration in hours
FTA - Fault Tree Analysis	D_i - Downtime per failure
O&M - Operation and Maintenance	P_F - Probability of failure
SCADA - Supervisory Control and Data Acquisition	μ - Repair rate per year
IGBT - Insulated-gate Bipolar Transistor	α - Scale parameter
HV - High Voltage	β - Shape parameter
GSU - Generator Step-Up	
DFIG - Doubly-fed Induction Generators	
PCB - Printed Circuit Board	
LCC - Life Cycle Cost	
OWPP - Offshore Wind Power Plant	
VSC - Voltage Source Converter	
XLPE - Extruded Cross-Linked Polyethylene	
SCOF - Self-Contained Oil-Filled	
HPOF - High-Pressure Oil-Filled	
GC - Gas Cables	
EPR - Ethylene Propylene Rubber	
PE - Voltage Source Converter	
MIND - Mass Impregnated Non-Draining	

truly global perspective on cable performance. The analysis not only scrutinizes different cable types but also considers their failure patterns within the unique operational challenges posed by marine and submarine applications.

- Expanding the scope of this review, a meticulous examination of circuit breaker failure rates has been conducted, with a focus on identifying the most critical components within the apparatus. The study draws upon data collected from 27 utilities, representing a diverse range of 13 countries. By analyzing failure patterns and performance metrics, this research aims to pinpoint the key vulnerabilities and challenges associated with circuit breakers.
- Finally a FTA of a wind farm is introduced, categorizing the system into several major subsystems: wind turbine, substations, HVAC/HVDC cables, and circuit breaker. These subsystems are then dissected and analyzed through the development of subtrees. The outcomes of this analysis offer insights for reliability allocation as well as operation and maintenance (O&M) management. However, it is noted that obtaining historical failure data for basic events is a prerequisite for implementing quantitative FTA. Yet, due to the limited number of wind farm samples, acquiring such data proves challenging.

Note that in the following, Section 2 clarifies basic concepts and definitions. Section 3 delves into the analysis of databases, providing insights into the overall reliability of wind turbines. Subsequent sections focus on specific components, including Section 4, which scrutinizes transformer reliability using dedicated databases, Section 5 exploring cable reliability, and Section 6 examining the reliability of circuit breakers. In Section 7, we conduct a comprehensive exploration of wind farm fault trees, unraveling intricate failure scenarios. A case study is conducted in Section 8 to calculate the failure and repair rates of a wind farm over 30 years. Finally, Section 9 offers a conclusion that synthesizes key findings, discusses implications, and suggests avenues for future research, providing a holistic view of the reliability landscape in wind farms.

2. Basic concepts and definitions in reliability

2.1. Approximating failure frequency and downtime duration

Reliability information, which encompasses failure rates and downtime data, is of utmost importance in the realms of system design, operation, maintenance, and performance evaluation. However, obtaining and sharing this data can often prove to be a challenging endeavor. This is due to the fact that the failure rate of a component depends on various factors. These factors can stay in a wide range from intrinsic characteristics to extrinsic causes; including physical characteristics of component's materials, operating conditions, design considerations, maintenance plans, operator and maintenance personnel expertise, climate conditions, etc. These factors can trigger a failure mechanism that may cause infant mortality failure, random chance failure or even wear-out failure. Therefore, the failure rate can decrease, stay constant or increase over the operation time. Thus, predicting the failure rate of a specific equipment is highly uncertain, even though a population of devices might have a failure behavior statistically closer to bathtub curve. Therefore, when there is a lack of data, failure rates are commonly assumed to be average. In power system engineering, this assumption is more realistic since the equipment is periodically maintained and the failure rate is kept close the average failure rate.

Decreasing failure rate, which is so-called infant mortality, usually occurs prematurely. Such failures can occur because of inadequate design, inferior material, poor workmanship, improper installation, etc., usually in new technologies. Prediction of the infant mortality failures is not straight forward. There is no mathematical or probabilistic method to predict those failures. In practice, various tests are performed during production of a component/system to prevent the infant mortality failures as much as possible. However, for reliability assessment, the historical data from operation of the similar cases are usually employed. The increasing failures, so-called wear-out failures occur due to the aging of materials operating long-term over continuous or higher stress conditions. These types of failures have intrinsic reasons attributed to the physical characteristics of materials, that can be predicted by using physical and mathematical models. For instance, in power electronics engineering, the failure rate of components such as capacitors and power devices are modeled using stress–strength analysis under given mission profiles. Depending on the operating conditions, material characteristics, and lifetime models, the end of life of a components can be predicted, thus, proper design can be performed to approach a desired level of lifetime under desired probability of failure.

In addition, random chance failures, which have a constant failure rate, are attributed to the extrinsic failures and usually due to the over-stressing the components/systems. These types of failure are also not easy to predict by mathematical or physical models as they have

extrinsic causes and they occur randomly. In the past, for instance in power electronic engineering, these failures had been modeled using generic data coming from field experiences or handbooks, and adapting them using acceleration factors. However, as the handbook data are not up-to-date and the given acceleration factors do not consider the physics of failures, those methods are no longer employed in power electronics engineering. The more acceptable approach for constant failure rate prediction is to use the recent historical data in the similar cases, i.e., for the same technology, similar operating conditions and similar weather conditions.

In power system engineering, estimating the failure rate and downtime of different components involves a systematic approach that begins with the collection and analysis of historical performance data, including condition monitoring systems, incident reports, supervisory control and data acquisition (SCADA) systems, and maintenance logs [18]. In reliability analyses, past occurrences are typically condensed and conveyed through two key metrics: the failure rate, representing the frequency of failures per component within a specific time frame, and the downtime per failure, indicating the duration a component remains non-productive due to a malfunction.

This review seeks to identify all available data sources that encompass failure and downtime statistics, collectively referred to as reliability statistics, for sub-assemblies within wind farms. The failure statistics can be expressed as either the annual calculated failure rates or the count of failures occurring within a specified time interval. In both scenarios, it is assumed that the time between failures follows an exponential distribution, and the failure process adheres to a homogeneous Poisson process characterized by a constant parameter, λ, which represents the failure rate [19–21].

Under the influence of this assumption, the calculation of failure rate, denoted as λ, takes the form of estimating the number of failures occurring per year, and the formulation is as follows [22,23]:

$$\lambda = \frac{\sum(\text{Number of units failed})}{\sum(\text{Number of units in service}) \times (\text{years in service})} \tag{1}$$

Given this assumption, the failure rate of subsystem s, denoted as λ_s and estimated as the count of failures per turbine per year, can be calculated from:

$$\lambda_s = \frac{\sum_{i=1}^{I} n_{s,i}/N_i}{\sum_{i=1}^{I} T_i/8760} \tag{2}$$

where

λ_s is the failure rate per subsystem per year.

I is the count of intervals for which data has been collected.

$n_{s,i}$ is the count of failures.

N_i is the number of turbines.

T_i is the overall duration in hours.

Thanks to Eq. (1), the failure rate of per turbine per year can be expressed as below [24,25]:

$$\lambda = \frac{\sum_{i=1}^{I} \sum_{k=1}^{K} n_{i,k}/N_i}{\sum_{i=1}^{I} T_i/8760} \tag{3}$$

where,

λ is the failure rate per turbine per year.

K is the quantity of subsystems.

Notice, $\sum_{i=1}^{I} \sum_{k=1}^{K} n_{i,k}/N_i$ represents the total number of failures across all time periods per turbine. In addition, $\sum_{i=1}^{I} T_i/8760$ is the summation of the duration in hours across all time periods, divided by the number of hours in a year.

In a manner akin to failure statistics, downtime statistics are commonly expressed either as the average calculated downtime per failure for each subassembly, denoted as D_i, or as the total downtime within a specific interval. Eq. (4) outlines the calculation of D_i.

$$D_i = \frac{\sum_{i=1}^{I} D_{s,i}}{\sum_{i=1}^{I} n_{s,i}} \tag{4}$$

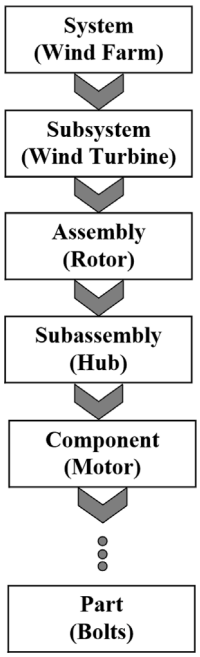


Fig. 1. Wind farm taxonomy.

where, $D_{s,i}$ represents the cumulative downtime resulting from failures of subsystem s during period i. In Eq. (4), the numerator signifies the aggregated downtime attributed to failures of subsystem s, while the denominator represents the total number of failures over the entire duration of a database.

2.2. Wind farm taxonomy

In this research, a standardized taxonomy is employed for reporting wind farm reliability data. The wind farm structure depicted in Fig. 1 serves as the basis for this common taxonomy. Table 1 illustrates the key elements of this taxonomy, facilitating a systematic survey of wind farm reliability data. This unified approach streamlines the process of estimating the associated failure rates and downtimes.

2.3. Data collection method

In this paper, data has been gathered from credible sources such as CIGRE, utility reports, and global surveys, with a focus on key components like wind turbines, cables, transformers, and protection units. To ensure representativeness, surveys from both onshore and offshore wind farms across various countries and climates were included, capturing a wide range of operational conditions. For example, transformer data was obtained from regions including Japan, Brazil, the UK, and CIGRE's global database. Potential biases, such as regional differences and specific equipment usage, were addressed by using multiple sources for each component and prioritizing peer-reviewed or industry-verified data that reflect long-term performance. These steps ensure the data is reliable and provides a comprehensive view of wind farm reliability.

3. Analysis of databases on the reliability of wind turbines

3.1. Wind turbine composition and configuration

In this paper, a standard classification system is employed to present the collected wind turbine reliability data. The wind turbine structure depicted in Fig. 2 serves as the basis for this classification. The standardized classification, as illustrated in Table 2, is adopted to facilitate

Table 1
Wind farm system structure and taxonomy employed in this review.

Subsystem	Subassembly
Wind turbine	Converter, Generator, Gearbox, Hydraulic system, Drive train, Rotor, Yaw system, Control system, Sensors, Structure and others
Transformer	Bushing and insulator, Coils, Mechanical relay, Tank, Cooling system, Core, Tap changer, Conductor and others
Circuit breaker	Auxiliary relays, Mechanical part, Wiring and joints, Coils, and others
Transmission system	Cable, Termination and Joints

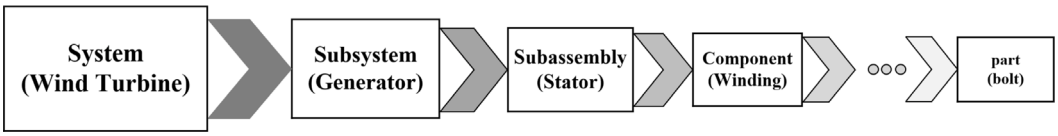


Fig. 2. Wind turbine structure.

Table 2
Wind turbine classification system [26].

Subsystem	Subassembly	Other Designations
Electrical	Converter	Frequency converter
	Transformer	–
	Electrical protection	Electrical components
Generator	Generator	–
Gearbox	Gearbox	System for converting speed
Hydraulic system	Hydraulic system	Hydraulics
Rotor	Pitch system	Pitch control
	Air brake	Rotor brake
	Blades	Rotor blades
	Hub	Hub unit
Yaw system	Yaw system	–
Drive train	Shafts and bearings	Low-speed, high-speed shafts
	Mechanical brake	Braking system
Nacelle	Nacelle	–
Structure	Tower	–
	Foundation	–
Sensors	Sensors	–
Other	Other	Other components

Table 3
Databases focusing on the reliability of wind turbines.

Name	Location	Years	Number of WTs	Reference
WMEP	Germany	1989–2006	1500	[1]
LWK	Germany	1993–2009	643	[2]
Windstat (1)	Denmark	1994–2004	2345	[36]
Windstat (2)	Germany	1995–2004	4285	[36]
Windstat (3)	Germany	2003–2012	24 000	[1]
Windstat (4)	Denmark	2003–2012	5000	[1]
Windstat (5)	Sweden	2003–2012	1200	[1]
Vindstat	Sweden	2000–2004	723	[37]
ReliaWind	Europe	2008–2011	350	[38]
CWEA	China	2010–2012	1311	[39]
CREW	USA	2011–2015	3000	[40]
CIRCE	Spain	2013	4300	[41]

the survey of wind turbine reliability data. This approach enables the estimation of associated failure rates and downtimes, fostering comparisons across different databases.

It is important to highlight that, within the scope of this section, the analysis is limited to the levels of system, subsystem, and subassembly as depicted in Fig. 2. A concrete instance of a subsystem is the electrical, comprising three subassemblies: converter, transformer, and electrical protection. Typically, a subassembly can be further broken down into smaller components. For instance, the pitch system encompasses the electric motors responsible for adjusting the blade pitch. Similarly, converters might involve discrete insulated-gate bipolar transistor (IGBT) modules, serving as bidirectional switches.

In the existing studies, there exists variability in the methods employed to dissect wind turbine data, leading to challenges in direct comparisons among different sources [27]. The classification presented in this paper derives from that utilized in the ReliaWind project, with minor adjustments [21,25,26]. The ReliaWind turbine classification system has been widely adopted in various studies concerning the reliability of wind turbine data, as noted in [27–30].

Additionally, it is important to note that a subassembly might be referred to by different names across various reliability sources. Table 2 presents the common names of wind turbine subsystems and subassemblies used in this paper and in the majority of reliability studies. The third column presents other designations that some studies in the literature may use to refer to a specific subassembly.

In Table 2, the classification labeled as *other* under subsystems is designated for equipment within the turbine that cannot be explicitly linked to any other specific category. For instance, certain surveys encompass data labeled as *unknown* or pertain to small components that may not be directly tied to the turbine's primary function, such as cabinets or lighting. All such components are collectively categorized under the *other* classification in this paper.

It is important to acknowledge that certain papers and reports in the literature focus on the reliability of either a single subassembly or the overall turbine performance, without delving into the detailed data of all subassemblies, as seen in [31–35]. These studies fall outside the scope of this survey because they lack a comprehensive turbine taxonomy, making it challenging to conduct a thorough comparison between their findings and other databases.

3.2. Databases for wind turbine reliability assessment

This review conducts a comprehensive review of wind turbine reliability data, encompassing 12 sources and around 48.6 thousand wind turbines from key countries in Europe, Asia, and America that hold significant roles in the wind energy sector. Table 3 provides details for each database, including the study's name, location, year, count of wind turbines, and references. Most of the data sources encompass a blend of a few hundred kilowatts (small-scale) to a few megawatts (medium-scale) turbines. The structure of the databases varies, with some countries like Denmark and Sweden having a single database, while others such as Germany feature multiple data sources.

3.2.1. WMEP database (Germany) [1]

The WMEP (Wissenschaftliches Mess und Evaluierungsprogramm) database is a comprehensive repository that offers detailed insights into the reliability and availability of wind turbines and their subassemblies.

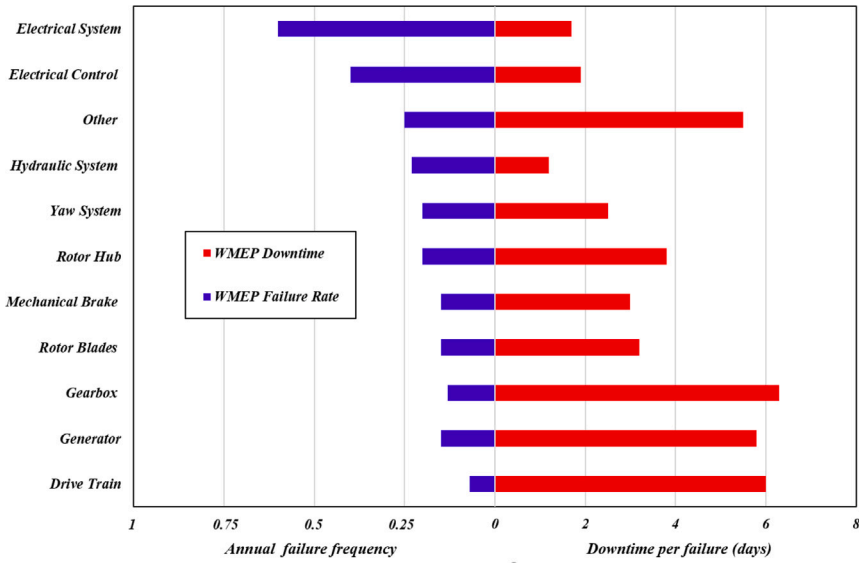


Fig. 3. Failure rate and downtime of subassemblies according to WMEP database [1].

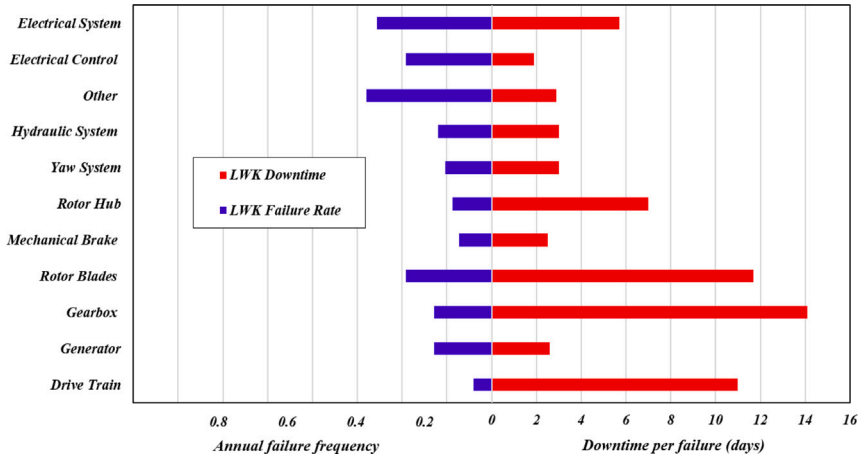


Fig. 4. Failure rate and downtime of subassemblies according to LWK database [2].

Initiated between 1989 and 2006, the program compiled an extensive dataset, encompassing 193,000 monthly operation reports and 64,000 incident reports derived from the operational experiences of 1,500 onshore wind turbines. This substantial collection of information spans a significant period, providing a valuable resource for understanding the performance and incidents related to these wind turbines over the specified timeframe. The frequency of failures for different subassemblies and typical downtimes per failure of the WMEP database are shown in Fig. 3 [1].

3.2.2. LWK database (Germany) [2]

The LWK-Database, initiated by the Schleswig-Holstein Chamber of Agriculture (LWK), serves as a data collection effort. Given that Schleswig-Holstein is the northernmost state in Germany and boasts the highest percentage of coastline, the wind turbines included in this database are exposed to relatively high wind speeds. Over the period between 1993 and 2009, a maximum of 643 wind turbines contributed data, encompassing information on both output and failures, to the LWK-Database. This dataset corresponds to a cumulative operational experience. The findings derived from this data collection were disseminated through an annual report. The frequency of failures for different subassemblies and typical downtimes per failure of the LWK database are shown in Fig. 4 [2].

3.2.3. Windstats database (Denmark, Germany and Sweden)

The Windstats newsletter or Windstats report has been in publication since 1988, serving as a commercial product currently owned by the Haymarket media group. Presently, the Windstats report is released on a quarterly basis, presenting performance data alongside news, reviews, and wind turbine production and operating data from turbines in Sweden, Denmark, and Germany. However, for this particular comparison, the original Windstats reports were unavailable. Therefore, existing analyses are considered, with the most recent publicly published results dating back to 2013 in Ref. [1]. However, due to the absence of information on absolute failure frequencies in this publication, it could not be fully incorporated into the comparison. As a result, data from a 2007 publication in Ref. [26] is included, offering a differentiation between Danish and German wind turbines. While the data collection has been ongoing since 1988, the analyzed dataset specifically spans the years 1994–2004. Noteworthy is the fact that the Ref. [26] solely provides information on failure rates, with no inclusion of downtime details for the subassemblies [26,36].

In addition, Ref. [1], provides some data from 2003 to 2012 as another Windstats newsletter. The data comes from 24000 wind turbines in Germany, 1200 wind turbines in Sweden, and 5000 wind turbines in Denmark. The failure frequencies of turbine subsystems for the years 2006, 2009, and 2012 are illustrated in Figs. 5, 6, and 7, respectively.

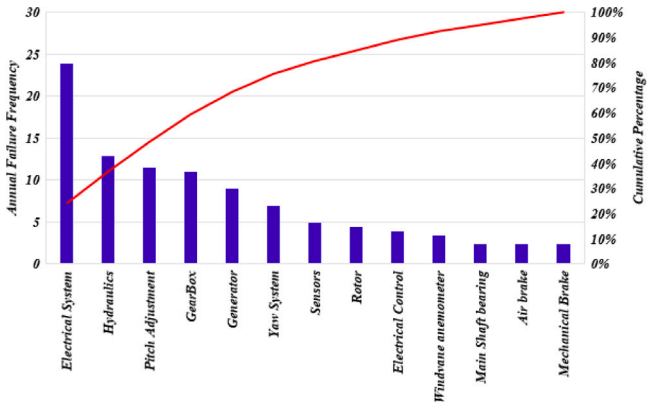


Fig. 5. The failure rate for each turbine subsystem on an annual basis in the year 2006 [2].

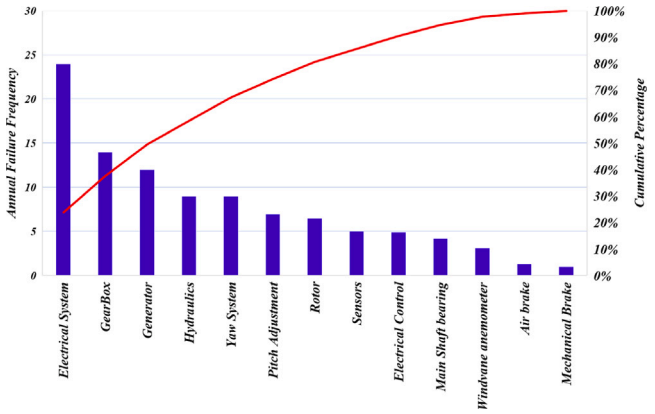


Fig. 6. The failure rate for each turbine subsystem on an annual basis in the year 2009 [2].

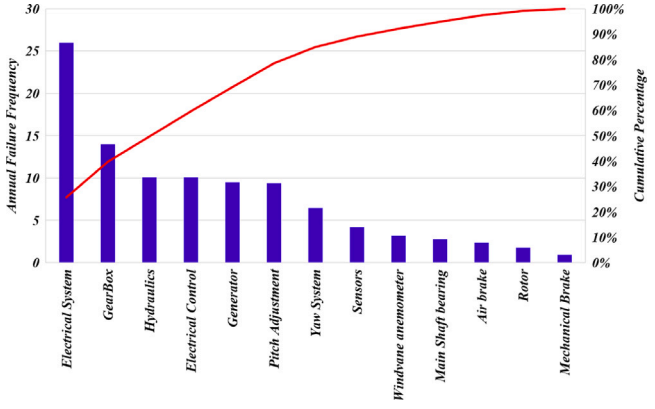


Fig. 7. The failure rate for each turbine subsystem on an annual basis in the year 2012 [2].

The consistent identification of the top four subsystems—electric systems, gearbox, hydraulics, and generator—reveals electric systems as the primary driver, maintaining its position as the foremost influencer. Simultaneously, the top seven drivers, responsible for over 80% of failure events annually, showcase dynamic shifts in their associated subsystems, indicating that the factors influencing system reliability undergo changes over time.

Furthermore, Figs. 8 and 9 depict the aggregated downtime per turbine subsystems for the periods 2003 to 2007 and 2008 to 2012, respectively. During both periods under consideration, the gearbox

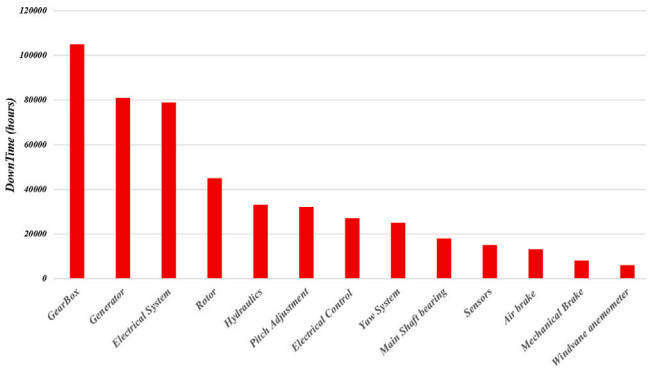


Fig. 8. The combined duration of turbine subsystem downtimes from 2003 to 2007 [36].

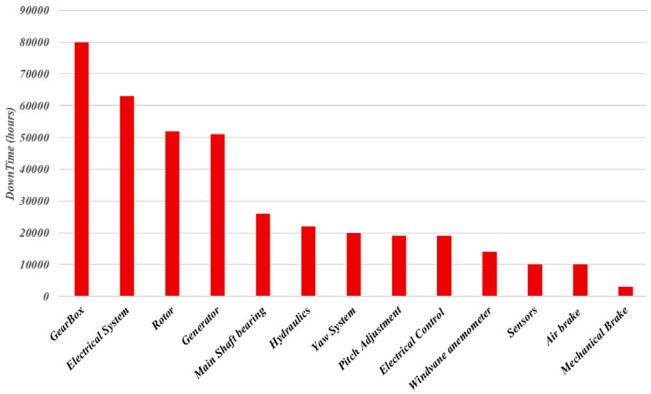


Fig. 9. The combined duration of turbine subsystem downtimes from 2008 to 2012 [36].

emerges as the predominant cause of downtime. In the more recent timeframe, a general decrease in downtime is observed across various subsystems compared to the earlier period. While the top four drivers remain the same, albeit with minor sequencing variations — gearbox maintaining its top position, followed by generator, electric systems, and rotor — the middle four drivers experience changes. Notice, in assessing the reliability of a composite power system, it is crucial to consider not only the frequency of failures but also the complexity and duration of interventions required during downtime. For instance, downtime is influenced by factors such as the type of component failure; while a failed IGBT may occur more frequently, its replacement can typically be executed swiftly. In contrast, interventions for components like gearboxes often entail longer durations due to their inherent complexity. In the older period, hydraulics, pitch adjustment, electric controls, and yaw were influential, whereas, in the recent period, main shaft/bearing, hydraulics, yaw system, and pitch adjustment take precedence.

3.2.4. Vindstat reliability databases (Sweden)

The assessment of the reliability of Swedish wind turbines relies on two main data sources. Vindstat issues an annual report on performance and availability [27], and the other one is a database on reported failures managed by Vattenfall. Both databases essentially cover the same wind turbine portfolio. The most recent publicly available reliability analysis based on this dataset was conducted by Ribrant and Bertling in 2007 [37], covering the years 2000–2004. The analysis encompasses a maximum of 723 wind turbines, while the complete database includes 786 wind turbines. It is noteworthy that the collection of detailed maintenance data are concluded in 2005 [42].

Fig. 10 illustrates the frequencies of failures and the corresponding downtime per failure for various components. The data suggests that

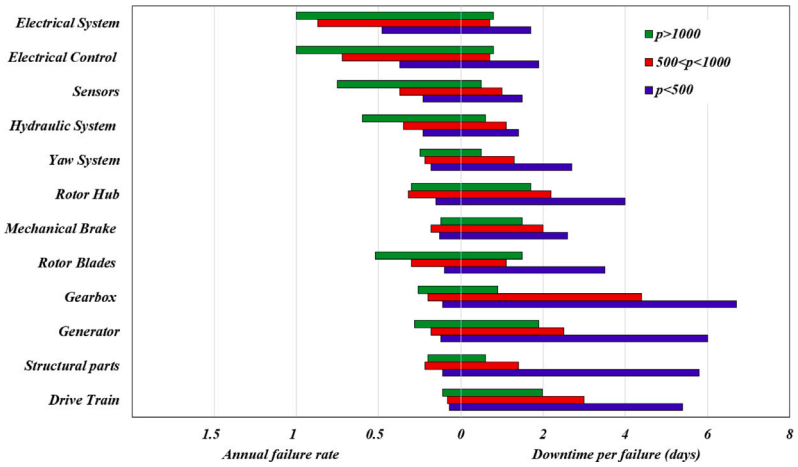


Fig. 10. Frequency of failures for different subassemblies and typical down time per failure based on Vindstat database [37].

failures in the electrical system, excluding the generator, are quite common. However, these failures tend to be addressed relatively quickly. This trend is particularly noticeable in larger turbines, where there is a higher incidence of electrical component failures. Additionally, it is likely easier to procure spare parts for these newer turbines, facilitating prompt repairs. In contrast, failures occur less frequently for the generator and the drive train. However, these are larger components, leading to longer repair times. The repair time for the gearbox on a large turbine appears relatively short in Fig. 10. However, it is essential to note that the study includes only a limited number of large turbines, potentially introducing bias to the results.

3.2.5. ReliaWind database (Europe)

The ReliaWind research project, funded by the European Union and conducted between 2008 and 2011, undertook the collection and analysis of SCADA-Data, fault/alarm logs, work orders, and service reports. The resulting ReliaWind Database encompasses 35,000 downtime events from approximately 350 wind turbines. Using this dataset, the project evaluated the failure rate and mean time to repair for the most crucial systems. However, due to confidentiality considerations, the reliability characteristics are presented not as absolute values but as relative (percentage) representations of the total failure rate/downtime. Consequently, while ReliaWind offers insights into the impact of individual systems/components, direct comparisons with other initiatives may not be feasible [38,43]. Fig. 11 depicts the normalized failure rates of subsystems and assemblies for turbines from various manufacturers in the database. This visual representation offers insights into the relative frequencies of failures across different components and subsystems, allowing for comparative analysis among turbines produced by multiple manufacturers. Furthermore, Fig. 12 illustrates the normalized hours lost per turbine per year due to failures in subsystems and assemblies for turbines from various manufacturers in the database. This visual representation provides an overview of the impact of failures on the operational time of turbines, allowing for comparative analysis among turbines produced by multiple manufacturers [38].

3.2.6. Chinese wind energy association (CWEA) database (China)

Ref. [39] discusses reliability and performance outcomes, drawing from the 2010–2012 Quality Report of China Wind Turbine Facilities by the Chinese Wind Energy Association (CWEA). In collaboration with 47 Chinese wind turbine manufacturers, component suppliers, and developers, CWEA gathered data on performance and failures spanning from 2010 to 2012. The dataset covered 111 wind turbines in 2010 and expanded to 640 turbines in 2012. The study highlights that most turbines are included in the analysis shortly after installation, which can impact the results of the survey: detecting early failures but not degradation and aging. Ref. [39] specifically details the total number of failures per technical system of the wind turbine, requiring the calculation of failure rates based on the provided data. However, the absence of specific details about the included portfolio, such as whether they are direct drive or gearbox turbines, introduces uncertainties to the findings. Furthermore, the study lacks information on the system structure used and does not provide details on the severity and downtime associated with the identified failures.

The distribution of component failure frequencies for Chinese wind turbines from 2010 to 2012 is shown in Fig. 13 and based on this figure frequency converters yield the highest failure rate among all of the components.

3.2.7. Continuous reliability enhancements for wind (CREW) database (USA)

The Continuous Reliability Enhancements for Wind Database and Analysis Program (CREW-Database) was launched by Sandia National Laboratories in 2007 with the aim of compiling operational data and status information on onshore wind turbines. The inaugural annual benchmark report was released in 2011 [27], and the most recent available results are from 2013 [40]. Notably, the data collection method involved the aggregation and analysis of alarm codes rather than maintenance reports. As a consequence, the results pertaining to event frequencies and average downtimes were not directly comparable to findings from other initiatives [44].

In 2016, the CREW program underwent a significant update, marked by Sandia's shift from collecting raw SCADA data to a curated collection of summarized data, facilitated by participating operators. These operators contribute master data concerning the included wind turbines, along with summarized SCADA-based availability data adhering to the definitions outlined in IEC 61400-26 [45]. Furthermore, operators provide summarized SCADA-based maintenance data and detailed information on individual maintenance records. Using this comprehensive dataset, the frequency of downtime events (how often) and the duration of downtime events (how long) are depicted in Fig. 14 with blue and red bars, respectively.

3.2.8. CIRCE, associated with the university of zaragoza in Spain

The study conducted at CIRCE-Universidad de Zaragoza compiled SCADA data and failure rates from a range of wind farms, turbine manufacturers, and types. The dataset includes information from about 4300 wind turbines across 230 wind farms, with rated capacities ranging from 300 kW to 3 MW. The data cover a period of approximately three years, likely around the year 2013. A comprehensive analysis of 7000 failure events and shutdowns was conducted, excluding those attributed to external causes. The data's organizational structure

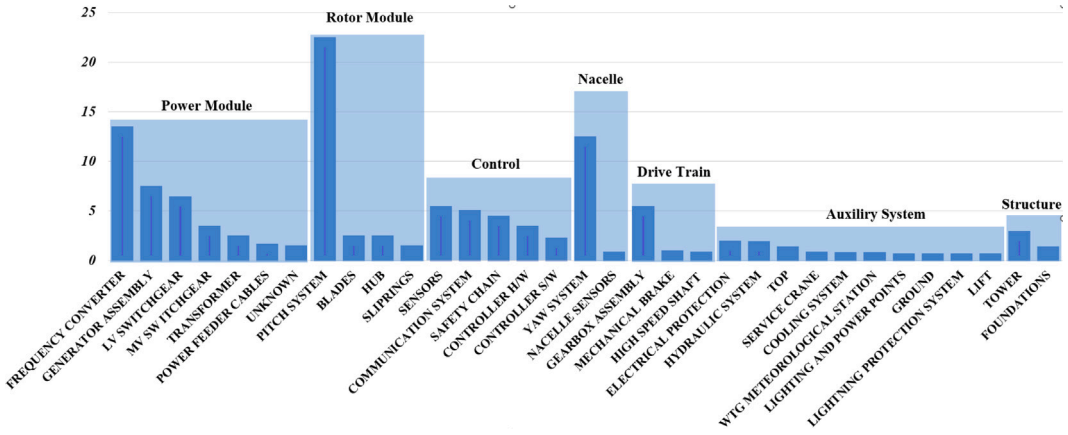


Fig. 11. Normalized failure rates of subassemblies according to ReliaWind database [38].

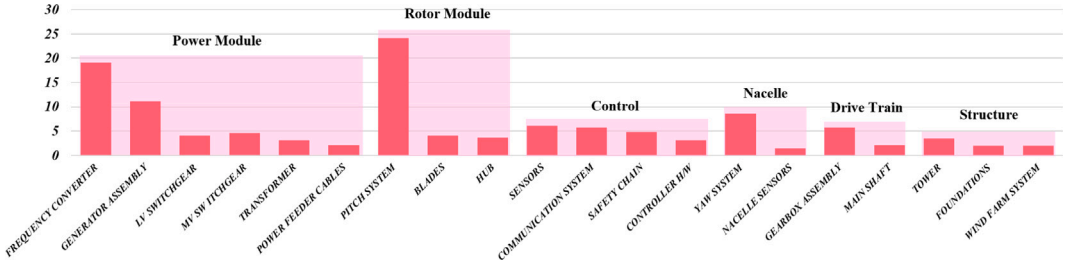


Fig. 12. Normalized hours lost of subassemblies according to ReliaWind database [38].

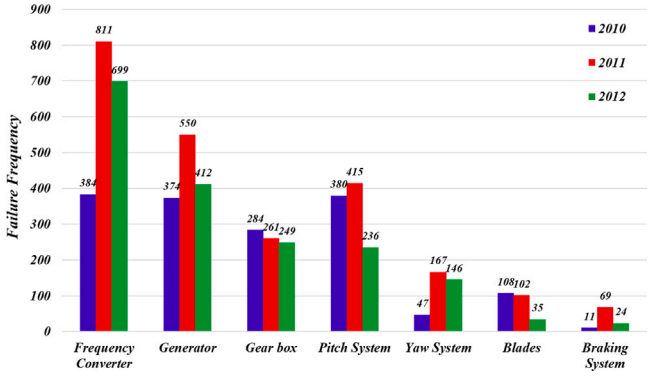


Fig. 13. Frequency distribution of component failures in Chinese wind turbines from 2010 to 2012 [39].

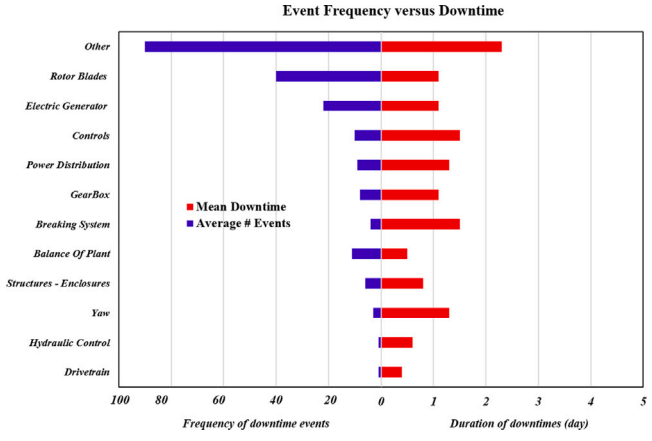


Fig. 14. Frequency of events and the duration of downtime based on the CREW database [40].

Table 4
Cumulative Failures and Downtimes per Turbine Annually for Various Wind Turbine Technologies in the CIRCE Study [41].

WT Technology	Failure/WT/Year	Down time/WT/Year
G < 1 MW	0.46	78.46 h
G ≥ 1 MW	0.52	44.51 h
DD	0.19	20.50 h

aligns with the system framework initially established by the ReliaWind project, tailored to suit the specific requirements of this project, making it distinctive. Failure rates are categorized by turbine concept (direct-drive or gearbox) and rated capacity (below or above 1 MW) [30,41]. According to this study, direct-drive turbines exhibited significantly fewer failures and less downtime compared to geared turbines. Table 4 illustrates the total failure rates per year and per turbine for each wind turbine type. The values are rounded and presented for geared wind turbines with a rated capacity below 1 MW (G < 1 MW), those equal to or above 1 MW (G ≥ 1 MW), and direct drive turbines (DD). Additionally, the average downtime per wind turbine per year and per failure is provided. It is worth noting that the obtained failure rates, in comparison to previous studies, fall within the lower range.

Figs. 15, 16, and 17 illustrate the findings concerning failure rates and downtimes per component categorized by turbine capacity (G < 1 MW, G ≥ 1 MW, and DD). Failure rates are presented as incidents per turbine per year, normalized to the total recorded failures, while downtimes are displayed as the component's impact on the overall downtime. The results indicate a distinct pattern for subsystems, highlighting that DD turbines were more prone to experiencing failures in power modules, control and communication systems compared to geared turbines. DD turbines showed minimal drive train failures, primarily limited to torque limiter issues, likely influenced by the absence of a gearbox—a major contributor to failure rates and downtimes. Concerning wind turbine assemblies, it is noteworthy that, irrespective of technology, the generator consistently stood out as a significant contributor to both overall failure rates and downtime.

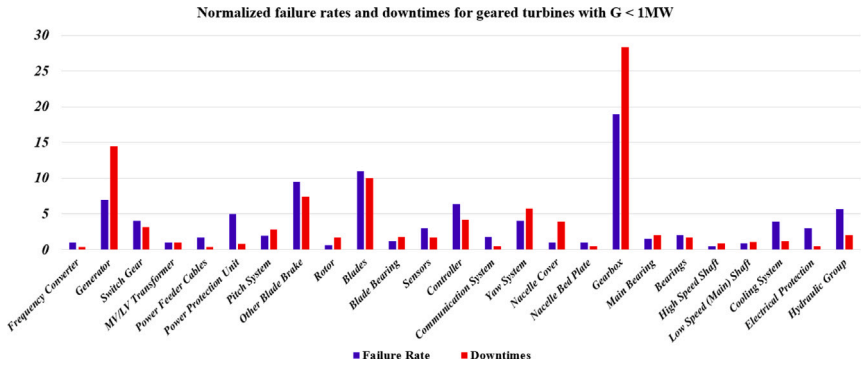


Fig. 15. Normalized failure rates and downtimes for geared turbines with G < 1 MW [41].

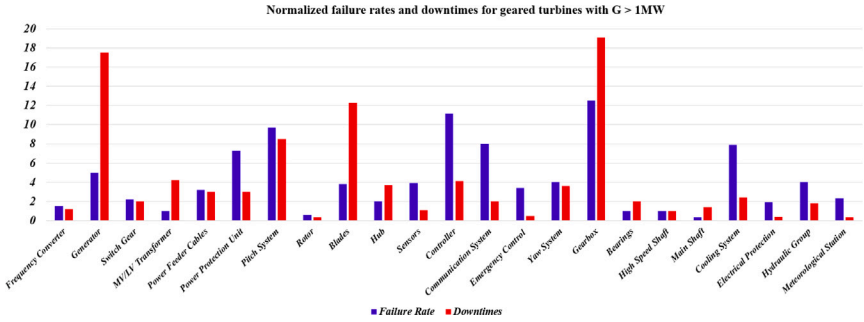


Fig. 16. Normalized failure rates and downtimes for geared turbines with G ≥ 1 MW [41].

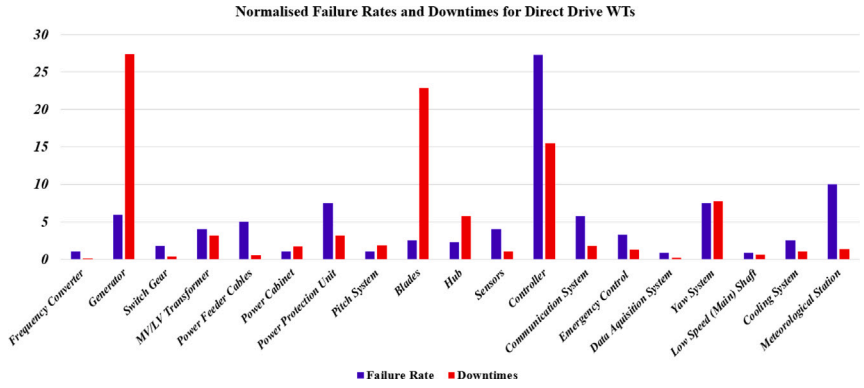


Fig. 17. Normalized failure rates and downtimes for direct drive wind turbines [41].

4. Transformer reliability databases

There is a limited body of publicly available literature discussing the failure statistics of transformers. Some countries and utilities are conducting internal surveys, and established databases in countries like Australia, Brazil, Canada, Germany, and Japan exist. However, the results and methodologies of these surveys are not publicly disclosed due to confidentiality and competition concerns among utilities. Another contributing factor to the scarcity of information may be the industry's tendency to rely on the comprehensive international survey as conducted in [22] rather than conducting individual surveys.

4.1. Transformer reliability database in Japan [3,4]

In the context of the current trend toward deregulation in the power market, Japanese power utilities are increasingly compelled to maximize the utilization of power facilities throughout their operational lifespan, minimizing the occurrence of unplanned outages. Utilities aim to acquire effective maintenance strategies by leveraging real-time data analysis for the proactive planning of preventive maintenance and potential replacement programs. As a result, comprehensive investigations have been conducted to gather diverse types of foundational information, including data on failures and issues related to power facilities such as transformers. The outputs of this research have been published in [3,4].

The administrative entity overseeing the committee is the Electric Technology Research Association of Japan, comprising primarily of various organizations including 11 power utilities, electrical manufacturers, universities, the Federation of Electrical Power Companies, the Central Research Institute of Electric Power Industry, and the Japan Electrical Manufacturers' Association. Table 5 presents the number of transformers installed by 10 utilities in Japan, distinguishing between those with a voltage capacity of up to 77 kV and those exceeding 77 kV. In addition, Table 6 presents both the count and the percentage rate of failures, where the average failure rate over the span of 10 years is noted as 0.09%. Notably, nearly 30% of these failures result from the malfunction of deteriorated mechanical relays, contributing to instances categorized as failures due to automatic shutdown, even

Table 5
The quantity of transformers units installed at 77kV and higher in Japan.

	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
Up to 77 kv	11 450	11 735	11 944	12 166	12 334	12 531	12 525	12 669	12 756	12 831
110 to 154 kv	851	872	884	886	891	897	905	920	930	932
187 to 275 kv	654	682	697	715	731	741	755	784	789	804
500 kv	206	224	230	246	262	276	268	273	275	279
Total	13 161	13 513	13 755	14 013	14 218	14 445	14 453	14 646	14 750	14 876

Table 6
Quantity of failures (rate of occurrences of failures per year) of transformers in Japan.

	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	Average
Up to 77 kv	15 (0.13)	11 (0.08)	20 (0.17)	10 (0.08)	12 (0.10)	10 (0.08)	3 (0.02)	11 (0.09)	7 (0.05)	3 (0.02)	102 (0.08)
110 to 154 kv	2 (0.24)	0 (0.00)	1 (0.11)	1 (0.11)	3 (0.34)	1 (0.11)	0 (0.00)	0 (0.00)	4 (0.43)	3 (0.32)	15 (0.17)
187 to 275 kv	0 (0.00)	0 (0.00)	0 (0.00)	1 (0.14)	0 (0.00)	4 (0.54)	0 (0.00)	2 (0.26)	2 (0.25)	2 (0.25)	11 (0.15)
500 kv	1 (0.49)	0 (0.00)	0 (0.00)	0 (0.00)	0 (0.00)	1 (0.36)	1 (0.37)	0 (0.00)	1 (0.36)	0 (0.00)	4 (0.16)
Total	18 (0.14)	11 (0.08)	21 (0.15)	12 (0.09)	15 (0.11)	16 (0.11)	4 (0.03)	13 (0.09)	14 (0.09)	8 (0.05)	132 (0.09)

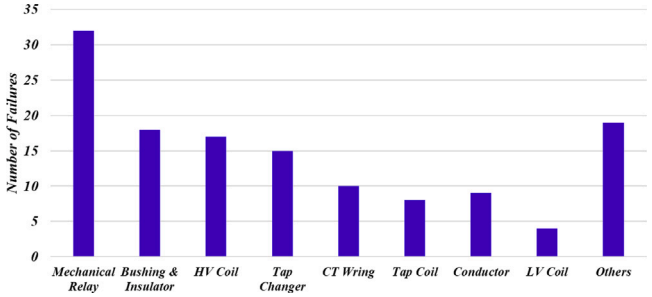


Fig. 18. The quantity of failures attributed to structural components of transformers in Japan [3,4].

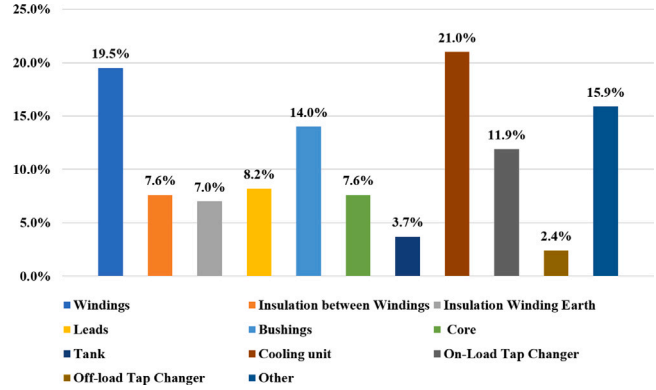


Fig. 19. Quantity of failures attributed to structural components of transformers in Brazil [5].

when the transformer remains undamaged. Although the population of analyzed transformers is different in each category, it is crucial to acknowledge that over 77% of these failures occurred within the voltage class up to 77 kV. Furthermore, transformers with voltages exceeding 100 kV exhibited a higher failure rate of 0.16%.

Fig. 18 illustrates the count of failures attributed to structural components from 1992 to 2001. Notably, high-voltage (HV) coil, bushing, and tap changer contribute significantly to the overall number of failures in transformer main components. It is noteworthy that mechanical relay operations, as indicated for reference, were primarily linked to malfunctions resulting from the deterioration of mechanical switches.

4.2. Transformer reliability database in Brazil [5]

Following the deregulation of the electrical market in Brazil, ABRATE, the Brazilian Power Transmission Utilities Association, established a dedicated working group tasked with compiling statistics on transformer performance. The association mandated that all member utilities provide data on transformer failures. The association publishes a comprehensive report biennially, although it is worth mentioning that only a limited number of utilities have agreed to share their data and insights for this initiative.

A questionnaire modeled after the Cigre WG 12-05 survey was meticulously crafted and distributed to various Brazilian utilities. Thirteen of the most significant utilities actively participated in the survey, contributing data from 3198 transformers and reactors. This dataset represents a substantial portion, exceeding 70%, of the installed capacity within Brazil. For detailed insights, a breakdown of the gathered data, categorized by voltage level, is presented in Table 7 [5].

Furthermore, Fig. 19 illustrates failures categorized by the involved components. Based on the data presented in this figure, off-load tap changer, winding, bushing, and on-load tap changer exhibit the highest rates of failure in transformers.

4.3. Transformer reliability database in United Kingdom [6]

An examination of significant failures in large generator and transmission transformers in the United Kingdom from 2003 to 2009 has been conducted. The survey encompassed large generator and transmission transformers, each with a capacity exceeding 100 MVA, predominantly linked to the 275 and 400 kV UK grid system. These transformers include around 800 transmission units with voltages ranging from 400/275 to 275/33 kV and ratings from 100 to 1,000 MVA, primarily functioning as auto-transformers. The majority of these units are over 40 years old. Given the overall low failure rates observed in transmission transformers, there was no deemed necessity to further subdivide these units into additional subgroups.

A compilation of 32 significant transformer failures that occurred between 2003 and 2009 has been documented. This compilation encompasses both in-service failures necessitating transformer replacement and instances where transformers were taken out of service preemptively due to concerns about their condition, constituting approximately one-third of the total incidents. The calculated failure rates are presented in the Table 8 based on Ref. [6].

The breakdown of causes for failures is depicted in Fig. 20. According to the information depicted in this figure, common causes of failure in transformers include aged insulation, manufacturing issues, over voltage, and tap-changer failures. It should be mentioned that, Fig. 20 provides a detailed breakdown of the causes behind 32 notable transformer failures that occurred between 2003 and 2009. It shows how these failures are distributed among various causes. For instance, insulation aging is responsible for 8 of the 32 failures.

Table 7
Number of units, number of failures and percentage of failures of transformers in Brazil [5].

Age	U ≥ 138 kV and e≤ 230 kV			U ≥ 230 kV and e≤ 345 kV			U ≥ 345 kV and e≤ 500 kV			U ≥ 500 kV			Total		
	# of units	# of failures	% failures	# of units	# of failures	% failures	# of units	# of failures	% failures	# of units	# of failures	% failures	# of units	# of failures	% failures
0–5 Years	150	2	1.33	344	8	2.33	142	6	4.23	167	3	1.80	803	19	2.37
6–10 Years	336	1	0.30	462	3	0.65	282	3	1.06	678	9	1.33	1758	16	0.91
≥11 Years	3618	18	0.50	4020	14	0.35	2670	21	0.79	2640	15	0.57	12948	68	0.53
Total	4104	21	0.51	4826	25	0.52	3094	30	0.97	3485	27	0.77	15509	103	0.66

Table 8
Number of units, number of failures and percentage of failures of transformers in United kingdom [6].

Transformer type	Subgroup	Population	# of failures	Failure rate %
GSU	G1	62	11	2.5
	G2	28	3	1.5
	G3	114	4	0.5
Transmission	T	800	14	0.3

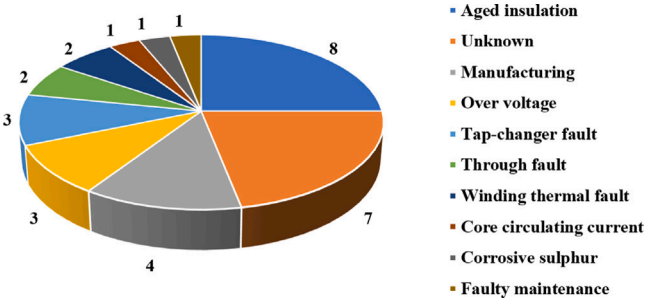


Fig. 20. Causes of transformer failures in the United Kingdom [6].

4.4. CIGRE transformer reliability database [6]

CIGRE compiled a dataset of 964 significant failures that occurred between 1996 and 2010, covering a total transformer operation time of 167,459 years. The data collection took place between 2010 and 2012. This data was provided by 56 utilities representing 21 countries. The manufacturing years of the units in question ranged from the 1950 to 2009. The reference periods, defined in the reliability questionnaire, varied in duration from 3 to 11 years, signifying the timeframe in which the documented failures occurred. Tables 9, 10, and 11 display the participating utilities along with their contributed transformer-years [6]. Since the operational transformer count was available for only one year, the total transformer-years (population per utility) were computed based on the assumption that the number of transformers in operation remained constant throughout the reference period.

The examined population of substation transformers amounted to a significant 150,072 transformer-years, as indicated in Tables 9, 10, and 11. This figure was nearly four times greater than the population in Bossi [22], which stood at 40,547 transformer-years. In contrast, the population of generator step-up (GSU) transformers was notably lower, totaling 17,387 transformer-years. Table 12 illustrates the investigated population and failure rates of GSU transformers. Moreover, the failure rates of GSU transformers are depicted in Fig. 21.

Fig. 22 illustrates the location of failures based on the application in units with voltages of 100 kV and above. According to the data shown in this illustration, the components with the highest rates of failure in transformers are HV winding, LV winding, HV bushing, HV lead exit, and tap changer.

5. Cable reliability databases

Over the past 25 years, substantial amounts of land and submarine cables, along with their corresponding accessories, have been installed. Concurrently, the technology and laying techniques associated with these cables have matured and undergone evolution. CIGRE Working Group B1.57 recently conducted a survey with the specific goal of gathering and analyzing data related to the installed quantities of land and submarine cable systems rated at 60 kV and above [46]. The survey encompassed an examination of the service experience and performance of existing land and submarine cable systems. Additionally, the collected data was utilized to identify and analyze trends in cable and accessory technologies.

The land cable survey received responses from 59 utility entities spanning 19 diverse countries, including Argentina, the United States of America, Australia, Austria, South Africa, Belgium, Brazil, The Netherlands, Denmark, Finland, Switzerland, Germany, Greece, Sweden, Italy, Japan, Spain, Malaysia and Norway. On the other hand, the submarine cable questionnaire garnered 21 responses from 14 different countries. The participating countries include Australia, Brazil, the United States of America, Denmark, Finland, Switzerland, France, Germany, Sweden, Greece, Italy, Spain, Japan, and Norway. The gathered data has undergone a thorough evaluation, scrutinizing both its quality and quantity. Through this assessment, and by comparing it to data from previous surveys, it is deemed representative and, consequently, serves as a valid foundation for updated reporting. Rigorous checks and validation procedures have been conducted to ensure the accuracy and reliability of the collected information.

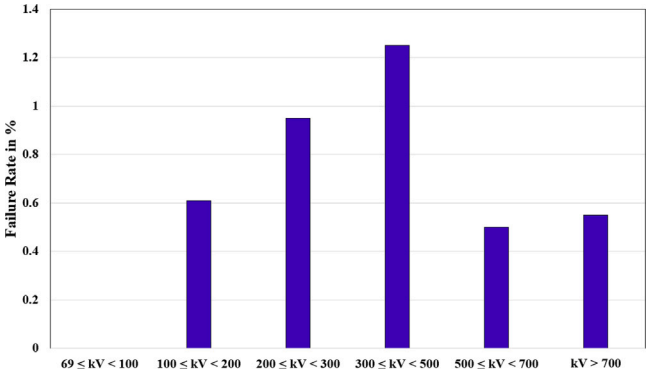


Fig. 21. Failure rates of GSU transformers in CIGRE transformer database [6].

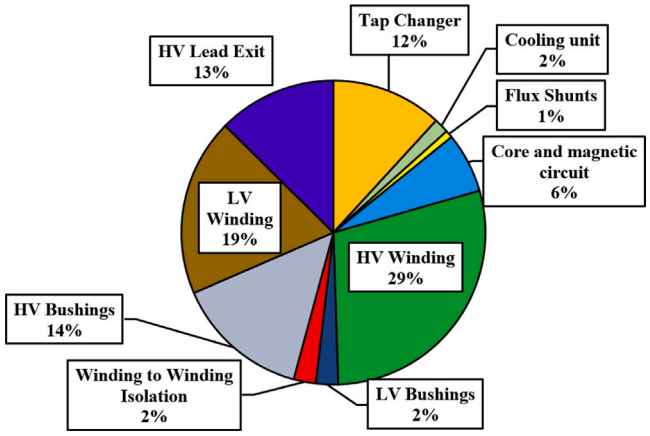


Fig. 22. Analysis of failure locations in GSU transformers with voltages $U \geq 100$ kV.

Table 9
Investigated European population of transformers in CIGRE database [6].

Country/Utility	Transformer-Years	Country/Utility	Transformer-Years	Country/Utility	Transformer-Years
Germany 1	3350	Austria 7	561	UK 4	2628
Germany 2	4730	Suisse 1	561	UK 5	1683
Germany 3	1848	Suisse 2	114	Netherlands 1	1947
Germany 4	160	Suisse 3	407	Italy 1	5670
Germany 5	85	Suisse 4	825	Ireland 1	3190
Germany 6	1990	Suisse 5	462	France 1	1368
Germany 7	1040	Suisse 6	1771	France 2	7416
Austria 1	1793	Czech 1	555	Denmark 1	781
Austria 2	407	Spain 1	8745	Denmark 2	440
Austria 3	1221	Croatia 1	950	Poland 1	2436
Austria 4	704	UK 1	6993	Hungary 1	1285
Austria 5	616	UK 2	1582	**Europe Total**	**71766 (42.86%)**
Austria 6	1408	UK 3	44		

Table 10
Investigated Australian population of transformers in CIGRE database [6].

Country/Utility	Transformer-Years	Country/Utility	Transformer-Years	Country/Utility	Transformer-Years
New Zealand 1	275	Australia 2	3289	Australia 7	77
New Zealand 2	286	Australia 3	88	Australia 8	1309
New Zealand 3	341	Australia 4	2244	Australia 9	2475
New Zealand 4	3399	Australia 5	594	**Australia Total**	**15774 (9.42%)**
Australia 1	1309	Australia 6	88		

Table 11
Rest of the world investigated population of transformers in CIGRE database [6].

Country/Utility	Transformer-Years	Country/Utility	Transformer-Years	Country/Utility	Transformer-Years
Brazil 1	29,920 (**17.87%**)	India 1	7,996 (**7.77%**)	Korea 1	17,784 (**10.62%**)
USA 1	3,146 (**1.88%**)	Canada 1	15,210 (**9.08%**)	South Africa 1	5863 (**3.50%**)

Table 12
The population and failure rates of generator step-up transformers.

Population and failures	Highest System Voltage (kV)						
	69≤kV<100	100≤kV<200	200≤kV<300	300≤kV<500	500≤kV<700	kV≥700	All
Number of Utilities	3	17	20	13	1	1	26
Number of Transformers	14	320	455	673	167	74	1703
Transformer-Years	153	3278	4639	6740	1837	740	17 387
Major Failures	0	20	43	89	9	4	165
Failure rate	**0.00%**	**0.61%**	**0.93%**	**1.32%**	**0.49%**	**0.54%**	**0.95%**

Table 13
Total length of AC land and submarine cables measured in kilometers [46].

Name/year of survey Voltage Range (kV)	CIGRE B1-57 2019	CIGRE B1-10 2009	Electra 137 1991	Worldwide EHV 2006	CIGRE B1-07 2007	EU Commission 2003
60–109	14997 (310)	17978 (786)	3646 (–)	0 (–)	24859 (–)	0 (–)
110–219	10475 (968)	9483 (1887)	717 (–)	0 (–)	22406 (–)	16000 (–)
220–314	3138 (88)	4457 (535)	1591 (–)	2000 (–)	5555 (–)	1720 (–)
315–500	569 (164)	999 (489)	239 (–)	200 (–)	1586 (–)	120 (–)
>500	171 (0)	N/A (N/A)	2 (–)	0 (–)	0 (–)	0 (–)
Total	**29350 (1530)**	**32917 (3697)**	**6195 (–)**	**2200 (–)**	**54406 (–)**	**17840 (–)**

Tables 13 and 14 illustrate the quantities of AC and DC cables, both land and submarine, as reported in recent surveys, with a comparison to the results of the WG B1-57 land cable survey and the older WG B1-10 survey published in 2009 (CIGRE TB379) [47–49]. Notice, in tables, the numbers enclosed in parentheses indicate the length of the submarine cables. Significantly divergent data is evident, primarily attributed to variations in survey scope and the specific questions posed.

The documented failures can be categorized into three primary groups:

- **Internal failures:** This type of failure stems from issues within the conductor, insulation system, screen, over sheath, or armor are a result of radial water penetration. In addition, any failure linked to oil leakage in an oil-filled cable or complications in other auxiliary systems is also classified as an internal failure. Instances of failures that trace back to the installation period of the cable system, such as those arising from poor workmanship or damage incurred during installation, are also deemed as internal failures.
- **External failures:** These failures encompass situations where malfunctions are attributed to human activities such as construction, excavation, anchoring, trawling, or fishing activities affecting submarine cables. Additionally, failures arising from the physical parameters and conditions surrounding the cables, such

Table 14
Total length of DC land and submarine cables measured in kilometers [46].

Name/year of survey Voltage Range (kV)	CIGRE B1-57 (2019)	CIGRE B1-10 (2009)
60–109	0 (0)	265 (143)
110–219	96 (193)	315 (822)
220–314	158 (1587)	20 (835)
315–500	745 (1707)	158 (1567)
>500	45 (1081)	0 (0)
Total	**1045 (4567)**	**758 (3366)**

as thermal conditions, as well as abnormal system or operating conditions like lightning strikes and earthquakes, are also categorized as external failures.

- **Unknown failures:** In cases where the cause of the failure remains unidentified or when no additional failure data have been reported, the situation is categorized as instances where the failure cause is unknown or undisclosed.

Table 15 presents the failure rates for various AC land cable systems, calculated using the received data.

Notice, in Table 15, XLPE, SCOF, HPOF, GC, EPR, PE and MIND are different cable types and are the abbreviations of Extruded Cross-Linked Polyethylene, Self-Contained Oil-Filled, High-Pressure Oil-Filled, Gas cables, Ethylene Propylene Rubber, Extruded Low-density thermoplastic polyethylene, and Mass-Impregnated Non-Draining cables, respectively. In addition, this table reveals distinct patterns in the failure rates of various AC land cable systems. Specifically, SCOF cable systems exhibit a failure rate approximately twice that of XLPE cable systems, while HPOF and EPR cable systems fall within a range of four times the failure rate of XLPE cables. PE cable systems stand out with an exceptionally high failure rate, surpassing 20 times that of XLPE cable systems.

Notably, the population of DC land cables is insufficient, and the reported failures are too low to calculate valid failure rates for DC land cable systems.

In Table 16, the distribution of failures across various submarine cable types is presented in relation to the voltage level. The failure rate is computed solely per cable type, encompassing all voltage levels and origins of failures (internal, external, and unknown). The sensitivity of the failure rate is highlighted through the calculation of the failure rate for one additional failure and one fewer failure, compared to the reported number of failures for each cable type. This approach provides insights into the variability and impact of reported failure data on the calculated failure rates, emphasizing the importance of sufficient data for accurate assessments.

Due to the relatively limited representation of submarine cables in the survey, caution is advised when drawing conclusions regarding the failure rates for this category of cables. Nevertheless, discernible patterns emerge from the data. AC XLPE submarine cables exhibit a higher failure rate compared to AC SCOF cables. Generally, DC submarine cables demonstrate notably lower failure rates than their AC counterparts. Also, extruded DC submarine cables appear to have a considerably lower failure rate than MIND and SCOF DC submarine cables. It is crucial to note that all reported DC cables with extruded insulation were installed between 2006 and 2015, indicating that these extruded DC cables are relatively new in terms of their installation period.

6. Circuit breaker reliability databases

A circuit breaker is a device designed to either open or close a high-voltage circuit by manipulating contacts through a mechanized system. Typically, circuit breakers employ solenoids to initiate the movement of contacts, releasing energy to the mechanical mechanism. The energy necessary for operating the circuit breaker's contacts is sourced from various methods, such as springs, pneumatic pressure, hydraulic pressure, or capacitor/motor systems. These energy sources are usually charged using motors or rectifiers, ensuring the effective functioning of the circuit breaker in managing high-voltage circuits [50]. The circuit breaker operates in response to an external command, which can be initiated either by an external relay or an operator. This command activates the mechanism, leading to a change in the position of the contacts. The external command is systematically processed by the circuit breaker control circuit, ensuring the correct and reliable activation of the circuit breaker. The circuit breaker control circuit constitutes of the electrical system essential for ensuring that the circuit breaker responds appropriately, safely, and reliably to external commands. It encompasses operational features like operating coils and monitoring facilities, which include indicators for both condition and position, ensuring a comprehensive control and monitoring capabilities [51].

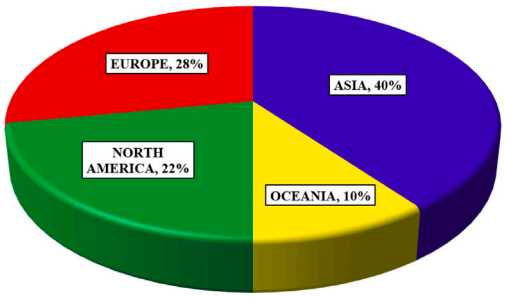


Fig. 23. Failure reports vs. geographical origin [52].

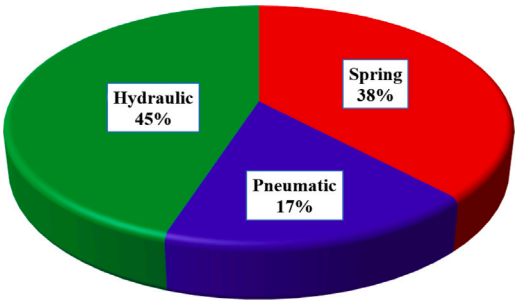


Fig. 24. Mechanisms for SF6 breakers [52].

CIGRE conducted a survey focused on the assessment of failure in circuit breaker control systems [52]. In this study, responses were gathered from 27 utilities representing 13 countries across various global regions for CIGRE's survey on circuit breaker control system failures. A total of 339 failure reports were deemed comprehensive enough to contribute to the study conducted by the CIGRE Working Group. The participating countries included USA, Germany, Japan, Netherlands, Romania, India, Switzerland, Australia, The Czech Republic, Poland, New Zealand, and Spain. The data's overall distribution is outlined in Fig. 23, categorized by geographic region.

Although variations exist in circuit breaker technology and the types of mechanisms identified within this investigation, nearly all newly installed equipment adopts $SF6$ technology. Consequently, when seeking enhancements in control system reliability for the future, the emphasis should be placed on $SF6$ technology.

In this survey, $SF6$ breakers are predominantly outfitted with hydraulic drives, as illustrated in Fig. 24. While this might be unexpected, it aligns with the findings of the second survey. The utilization of spring mechanisms has shown a consistent increase since 1994; however, it has not yet become the predominant type based on the data reported in this study.

Given the absence of information about the total population of breakers from which the reported control failures have been drawn, it

Table 15
Failure rates on the different types of AC land cables. All origins (Unit: failures/100 circuit-km-year).

Failure rate	All voltages	60–109 kV	110–219 kV	220–314 kV	315–499 kV	500 kV and above
XLPE	0.111 21538 km/204 faults	0.0939 12848 km/107 faults	0.108 6359 km/55 faults	0.269 1874 km/40 faults	0.102 1874 km/40 faults	0 114 km/0 faults
SCOF	0.224 6123 km/137 faults	0.0426 1885 km/8 faults	0.352 3105 km/109 faults	0.140 863 km/12 faults	0.376 213 km/8 faults	0 58 km/0 faults
HPOF	0.436 681 km/29 faults	1.45 148 km/21 faults	0 127 km/0 faults	0.209 394 km/8 faults	0 12 km/0 faults	- - km/- faults
GC	0.157 450 km/7 faults	0 9 km/0 faults	0.163 433 km/7 faults	0 7 km/0 faults	- - km/- faults	- - km/- faults
EPR	0.373 452 km/16 faults	0.567 75 km/4 faults	0.305 376 km/11 faults	- - km/- faults	0 1 km/0 faults	- - km/- faults
PE	2.649 106 km/28 faults	0.641 31 km/2 faults	3.49 75 km/26 faults	- - km/- faults	- - km/- faults	- - km/- faults

Table 16
Quantity and failure rates of submarine cables between 2006 and 2015.

Cable types	Voltage Range (kV)						Rate
	60 to 109	110 to 219	220 to 314	315 to 500	above 500	All voltages	Rate −1/+1 fault
AC – SCOF	38 km 0 fault	163 km 1 fault	8 km 0 fault	135 km 3 faults	- km - fault	344 km 4 faults	0.12 (0.092–0.15)
AC – HPOF	53 km 3 faults	37 km 0 fault	22 km 0 fault	- km - fault	- km - fault	112 km 3 faults	0.29 (0.19–0.38)
AC – XLPE	203 km 3 faults	768 km 2 faults	58 km 2 faults	29 km 0 fault	- km - fault	1059 km 7 faults	0.18 (0.15–0.20)
AC - EPR	15 km 0 fault	- km - fault	- km - fault	- km - fault	- km - fault	15 km 0 faults	0.0 (0.0–1.3)
DC - SCOF	- km - fault	- km - fault	656 km 1 fault	- km - fault	93 km 0 fault	749 km 1 faults	0.02 (0.0–0.039)
DC - MIND	-km - fault	- km - fault	681 km 3 faults	1388 km 4 faults	988 km 0 fault	3056 km 7 faults	0.029 (0.025–0.033)
DC – Extruded	- km - fault	193 km 0 fault	251 km 0 fault	319 km 0 faults	- km - fault	763 km 0 faults	0.0 (0.0–0.047)

is not feasible to ascertain an overall failure rate from this dataset. Consequently, a comparison with the outcomes of the 1994 survey cannot be conducted, and the data analysis is constrained to an examination of the failure caused by component. After conducting analysis, it was established that the control circuit experiences the highest failure rate, primarily due to its tendency to malfunction and fail to operate. In addition, unintended operations constitute only 13% of major failures. The initial analysis suggests the challenge of pinpointing a specific component as a major source of reliability issues. The evaluation indicates that six types of components significantly contribute to the overall number of failures, as illustrated in Fig. 25.

The six components, collectively responsible for 88% of all control failures, include:

- Auxiliary relays.
- Position auxiliary switch (mechanical parts).
- Coils.
- Energy limit supervision.
- Density monitoring (including associated piping).
- Wiring and joints.

7. Wind farm fault tree analysis

7.1. Exploring fault tree analysis

Fault tree analysis is a structured and visual technique applied across diverse industries to investigate the origins of unwanted events, breakdowns, or system weaknesses [53]. FTA was created in 1961 at Bell Laboratories by H. A. Watson, with the collaboration of M. A. Mearns. Its inception was motivated by the need to aid in the development of the US Air Force's Minuteman missile system [54]. Subsequently, this method found successful application in the comprehensive system analysis undertaken by David Haasl at the Boeing Company. Following its inception, FTA has been widely adopted and applied across diverse sectors, encompassing, but not limited to, automotive, aerospace, and nuclear industries [55–57]. FTA is a systematic, top-down approach that examines potential failure modes and their interconnections within complex systems. Its primary objective is to pinpoint and comprehend the fundamental reasons behind specific events or system malfunctions, typically within the realms of safety, reliability, and risk assessment [58,59]. In other words, the analysis outcomes reveal how the failures of various components or specific environmental conditions can come together to result in a system failure. A fault tree is a directed acyclic graph with gates and events as its two types of nodes. Fig. 26 demonstrates the configuration of a fault tree.

7.1.1. Events of a fault tree

An event represents an incident within the system, usually involving the malfunction of a subsystem, even down to the level of an individual component. Events can be categorized into two groups: basic events and intermediate events. Basic events that happen spontaneously are fundamental elements that cannot be broken down further, and they are typically characterized by their probability of failure, which is usually determined from failure statistics over time. Intermediate events that result from the occurrence of one or more other events are depicted

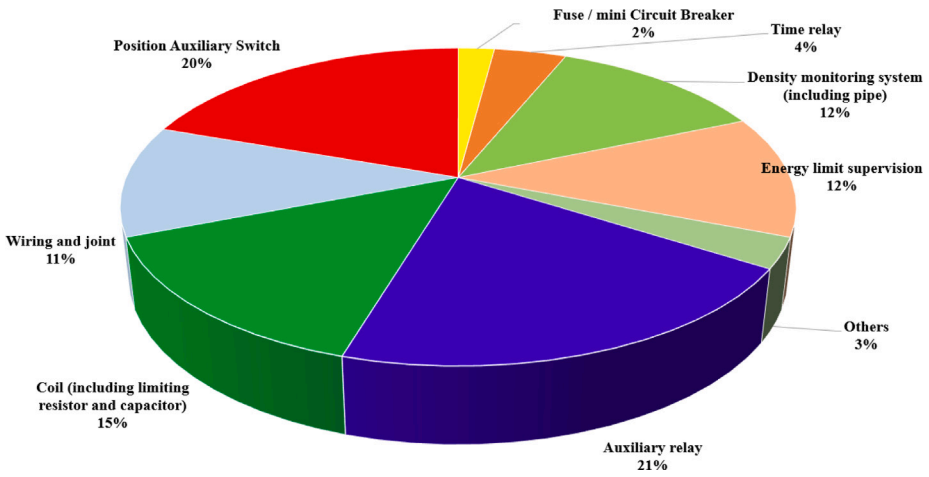


Fig. 25. Total failure reports vs components [52].

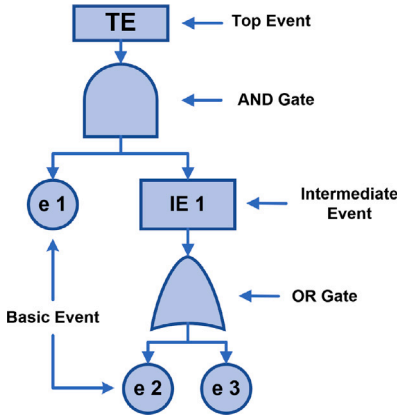


Fig. 26. Configuration of a fault tree.

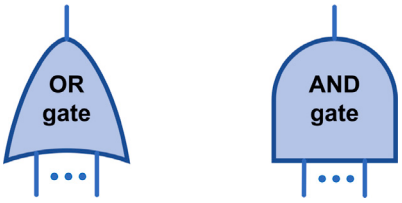


Fig. 27. Common gates in a fault tree.

by combining basic events and other intermediate events using logical gates, and they play a crucial role in illustrating the progression of failure within a system [60]. Intermediate events and failure investigations like the physics of failure can be valuable for documentation but typically do not affect or influence the analysis of the fault tree, and as such, they can be left out or omitted from the analysis. The event positioned at the highest point of the fault tree, referred to as the top event, is the event subject to analysis, serving as a representation of the system's failure that is under examination [61]. In other words, the top event is a catastrophic scenario that results in system failure, and its occurrence is explained in terms of the occurrence or non-occurrence of other events [62].

7.1.2. Gates of a fault tree

Gates symbolize the way failures spread throughout the system, illustrating how failures within the system can converge to result in a system failure. Each gate has a single output and one or more inputs.

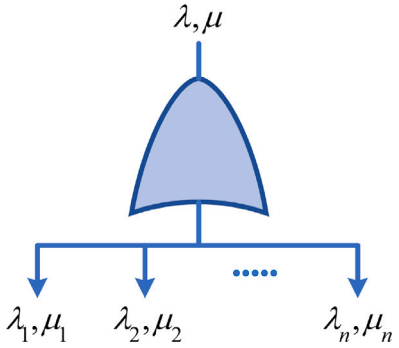


Fig. 28. A fault tree including an OR gate.

The gates utilized in a fault tree analysis include OR, AND, k/N gate, and INHIBIT gate. The following gates are frequently employed in fault trees. Visual representations of these gates are shown in Fig. 27 [63,64].

- **OR:** An OR gate of a fault tree is depicted in Fig. 28. An output event of an OR gate happens when any of the input events occur. The OR gate relates to a series system with n independent occurrences, where Eq. (5) can be used to determine the probability of failure. Utilizing the average failure rates and average downtimes of the input events, Eqs. (6) through (8) are applied to determine the output event of an OR gate.

$$P_F = 1 - \prod_{i=1}^{n}(1 - P_i) \tag{5}$$

$$\lambda = \sum_{i=1}^{n} \lambda_i \tag{6}$$

$$\mu = \sum_{i=1}^{n} \lambda_i \frac{\prod_{i=1}^{n} \mu_i}{\prod_{i=1}^{n}(\lambda_i + \mu_i) - \prod_{i=1}^{n} \mu_i} \tag{7}$$

$$P_F = \frac{\lambda}{\lambda + \mu} = 1 - \prod_{i=1}^{n} \frac{\mu_i}{\lambda_i + \mu_i} \tag{8}$$

- **AND:** An AND gate of a fault tree is depicted in Fig. 29. An output event happens when all of the input events occur. The AND gate is associated with a parallel system, and the probability of failure is determined by multiplying the probabilities of the n independent events, as described in Eq. (9). To compute the output event of an AND gate, Eqs. (10) to (12) are employed.

$$P_F = \prod_{i=1}^{n} P_i \tag{9}$$

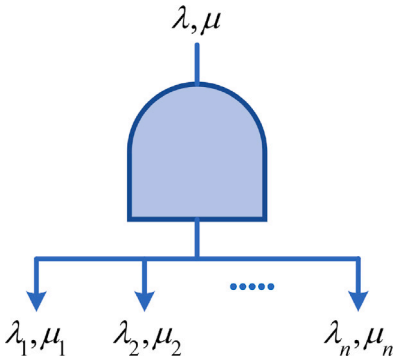


Fig. 29. A fault tree including an AND gate.

$$\mu = \sum_{i=1}^{n} \mu_i \tag{10}$$

$$\lambda = \sum_{i=1}^{n} \mu_i \frac{\prod_{i=1}^{n} \lambda_i}{\prod_{i=1}^{n} (\lambda_i + \mu_i) - \prod_{i=1}^{n} \lambda_i} \tag{11}$$

$$P_F = \frac{\lambda}{\lambda + \mu} = \prod_{i=1}^{n} \frac{\lambda_i}{\lambda_i + \mu_i} \tag{12}$$

Following the construction of a fault tree, the analyses of the model are conducted at a qualitative level and the other at a quantitative level. In qualitative analysis, the typical approach involves simplifying fault trees into minimal cut sets. These minimal cut sets are essentially a collection of the smallest combinations of basic events that are both necessary and sufficient to trigger the top event. In quantitative analysis, mathematical calculations are employed to determine the probability of the top event occurring and other quantitative reliability measures, like importance measures. These calculations are based on the failure rate or probability of individual system components. The outcomes of quantitative analysis provide analysts with insights into system reliability. Events with high importance measures values are regarded as critical elements, suggesting that primary inspection and maintenance tasks should be prioritized to ensure the system's safety [59].

In this review, a top-down approach was used to identify failure modes for each component, with fault trees constructed to trace their potential impact on the overall system. Failure rate data from multiple sources were utilized to quantify the risks and estimate how individual component failures affect the wind farm's overall reliability. However, it should be noted that FTA assumes independence between failure events, which may not always reflect real-world conditions, particularly in cases of common-cause failures. Additionally, the method simplifies complex failure scenarios by treating them as binary events (failure or no failure), which may not fully capture all the nuances of actual failures. Despite these limitations, FTA remains a useful tool for identifying failure pathways and prioritizing areas for reliability improvement.

In assessing the reliability and potential risks of a wind farm, a comprehensive fault tree analysis has been developed, encompassing various critical components. The analysis considers primary sources of failure, including wind turbine malfunctions, transformer failures, HVAC/HCDC system breakdowns, and circuit breaker failures. The interconnected nature of these elements is visually depicted in Fig. 30, providing a holistic view of potential fault scenarios within the wind farm. Furthermore, the subsequent sections will delve into a detailed examination of the fault tree analysis for each specific part, namely the wind turbines, substations including transformers, converters, filters and ..., HVAC/HCDC cables, and circuit breakers. By isolating and scrutinizing the potential failure modes and their interdependencies within these critical elements, a comprehensive understanding of the wind farm's reliability and vulnerability profile can be achieved. This focused approach allows for targeted interventions and maintenance strategies tailored to the unique challenges posed by each component, contributing to the overall resilience and operational stability of the wind farm.

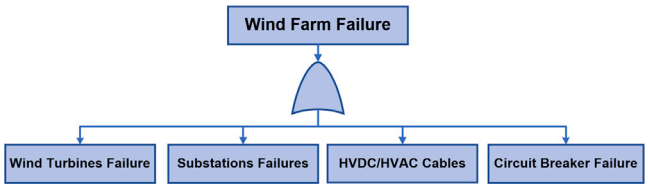


Fig. 30. Wind farm's fault tree.

7.2. Wind turbine system level fault tree

A wind turbine can be categorized into ten primary subsystems based on their respective functions. These include a generator, converter, transformer, control system, gearbox, yaw system, blade system, pitch and hydraulic system, and support structure. Various components of a wind turbine work together to efficiently harness wind energy and generate electricity. The aerodynamically designed blades capture kinetic energy from the wind, which drives the rotor to spin. The rotational energy is then transferred to the generator, where it is converted into electrical energy through electromagnetic induction. The converter adapts the electrical output for grid compatibility if required. Some turbines incorporate a gearbox to increase rotational speed before reaching the generator. The tower supports the rotor and nacelle at an optimal height for wind capture. Within the nacelle, control systems monitor and adjust turbine orientation and blade pitch for maximum energy capture and protection in varying wind conditions. This collaborative effort among the blades, rotor, generator, converter, gearbox, tower, nacelle, and control systems ensures the efficient and reliable generation of electricity from wind power. Any failure within these subsystems can result in malfunction of the entire system. This situation exemplifies a typical scenario of a series system, OR gate in the fault tree, which is illustrated in Fig. 31 and encompasses ten primary systems listed in Table 17 where s stands for subsystems.

Noticed, this study's data was gathered using a thorough method that combined information from existing literature with engineering expertise. Some data came from peer-reviewed papers and industry reports, capturing established trends and empirical findings. Engineering judgment was crucial, filling gaps and providing insights based on practical experience. This approach enhances the depth and comprehensiveness of the analysis.

In addition, the fault trees presented in this paper are based on the most probable failure events identified in current literature and industry practices. While they provide a solid foundation, they are not exhaustive. To fully address all potential failure modes, further refinement and validation would be necessary. These fault trees should be viewed as a preliminary framework, with additional work needed for complete real-world applicability.

Finally, due to insufficient data on repair costs or time for basic events in the fault trees across various sources, this paper does not include information on repair times.

7.2.1. Fault tree of a wind turbine generator

The generator subsystem in a wind turbine situated within the nacelle is a pivotal component responsible for converting the mechanical energy harnessed by the rotating rotor into electrical power. Based on [28,65], this equipment holds notable significance due to its elevated failure rates, downtime, and associated repair expenses. The wind turbine generator fails because of electrical failures, mechanical failures, environmental factor failures, or human error failures [66].

Table 17
Abbreviations and descriptions of a wind turbine different subsystems in Fig. 31.

Abbreviation	Description	Abbreviation	Description
S1	Generator failure	S6	Yaw system failure
S2	Converter failure	S7	Blade system failure
S3	Transformer failure	S8	Pitch and hydraulic system failure
S4	Control system failure	S9	Support structures failure
S5	Gearbox failure		

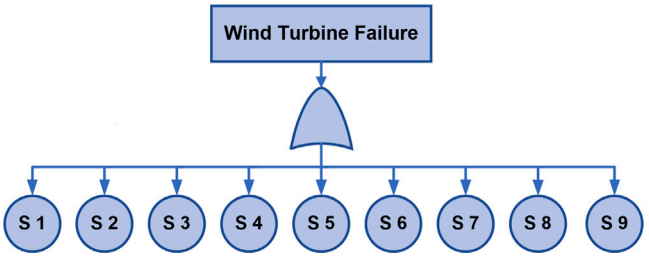


Fig. 31. Wind turbine's fault tree.

Table 18
Intermediate events of a wind turbine generator fault tree.

Intermediate event	Codes
Environmental factor	g1
Mechanical	g2
Electrical	g3
Human error	g4
Bearing Failure	g5
Stator Winding Failure	g6
Rotor Winding Failure	g7

Table 19
Basic failure events of a wind turbine generator fault tree [19,71–73].

Basic events	Codes	λ (f/yr)
Lightning strikes	e1	0.003504
Freezing Condition	e2	0.006132
Shaft failure	e3	0.004818
Voltage regulator failure	e4	0.006745
Control system malfunction	e5	0.004818
Installation error	e6	0.004380
Maintenance error	e7	0.004380
Misalignment (Asymmetry)	e8	0.001927
Abnormal vibration	e9	0.004818
Material fatigue	e10	0.017344
Insulation breakdown	e11	0.004818
Short circuit	e12	0.015417
Insulation breakdown	e13	0.004818
Short circuit	e14	0.011563
Rotor imbalance	e15	0.001927

It is worth mentioning that lightning strikes as well as freezing conditions can cause environmental factor failure. Furthermore, electrical failures originate from failures in stator winding, rotor winding, voltage regulator, or control system. In addition, shaft failure, misalignment (asymmetry), abnormal vibration, and material fatigue account for the largest share of mechanical failures [67,68]. Finally, human error failures happen because of installation errors and maintenance errors [67–70]. The fault tree of a wind turbine generator is presented in Fig. 32 and Table 18 and Table 19 list intermediate and basic failure events, respectively.

7.2.2. Fault tree of a wind turbine converter

Power converters play a pivotal role in nearly all contemporary wind turbines. Positioned between the generator and the wind turbine transformer, they facilitate variable-speed operation and guarantee that the turbine operates in accordance with grid regulations [74]. In the context of wind turbines, different generator-converter configurations are commonly utilized. These include systems with doubly-fed induction generators (DFIG) equipped with partially rated converters, setups where squirrel-cage induction generators are combined with full-scale converters, and configurations where permanent magnet or electrically excited synchronous generators (PMSG, EESG) are paired with converters rated at their full capacity [35]. Converter's failure rate is high and it is not restricted to particular manufacturers or locations; instead, it is a global concern affecting various wind turbine models, as substantiated by numerous research findings and strongly supported by the collective experience of wind turbine operators [25]. As shown in Fig. 33, printed circuit board (PCB) failure, terminal damages, IGBT module failure, capacitor failure, and control failure are all contributing factors to the occurrence of converter failures [75]. IGBT module failure originates from die (chip) failure or package (bond wire + solder joint) failure. Each of these items can come from several intermediate and basic events demonstrated in Fig. 33, Tables 20 and 21 [76]. Although there is a lack of information on the average duration for repairing various components of a wind turbine converter, the repair is fairly quick because they are usually replaced.

Table 20
Intermediate events of a wind turbine converter fault tree.

Intermediate event	Codes
IGBT module failure	g1
Package failure	g2
Solder joint failure	g3

Table 21
Basic failure events of a wind turbine converter fault tree [25,71,74–76].

Basic events	Codes	λ (f/yr)
PCB failure	e1	0.046515
Terminal damages	e2	0.005431
Capacitor failure	e3	0.053874
Control failure	e4	0.012526
Die (chip) failure	e5	0.018396
Bond wire failure	e6	0.019359
Base plate solder joint failure	e7	0.015505
Die attached solder joint failure	e8	0.007708

7.2.3. Fault tree of a wind turbine transformer

Wind turbine transformers can be situated in various locations, including adjacent to the turbine but external to it, at the base of the tower, or within the nacelle. The optimal location for efficiency is in proximity to the power source, whenever feasible. The main function of the wind turbine generator step-up (GSU) transformer is to raise the low voltage output initially generated by the wind turbine to the higher voltage level [77]. Failures of GSU transformers are identified as a primary cause of extended downtime in wind turbines. Addressing such failures involves costly, complex, and time-consuming replacement procedures, particularly in adverse weather conditions. Therefore, in certain instances, discontinuing their use may be a preferred course of action [78]. Wind turbine GSU transformer faults can occur due

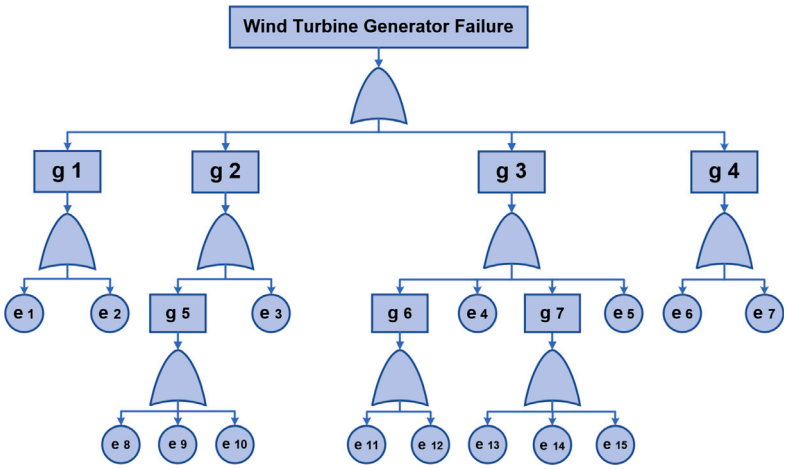


Fig. 32. Fault tree of a wind turbine generator.

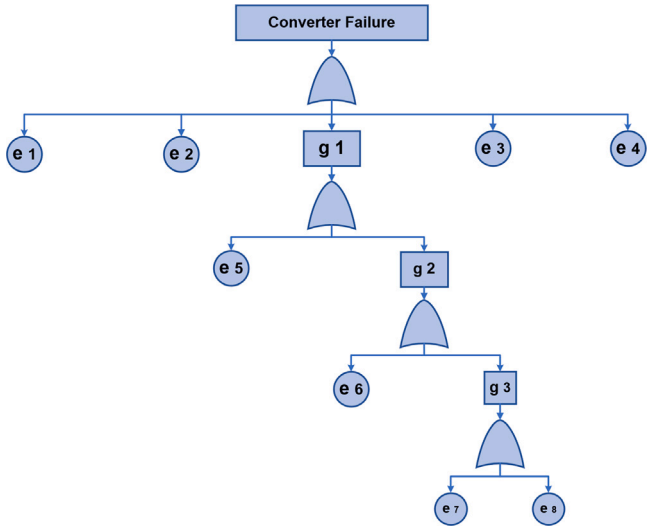


Fig. 33. Fault tree of a wind turbine converter.

Table 22
Intermediate events of a wind turbine GSU transformer fault tree.

Intermediate event	Codes
Bushing failure	g1
Winding fails to induce electricity	g2
Core fails to be path of magnetic flux	g3
Paper insulation fails	g4
Oil insulation fails	g5
Oil level gauge fails to initiate trip	g6

to a range of factors, including bushing failure, winding failure, tank failure, and core failure [79]. Failures in wind turbine GSU transformers can be triggered by various factors. Bushing failure occurs because of failure on conductive and insulation parts. Failures in both paper and oil insulation, as well as coil element failures sow the seeds of winding failure. Additionally, issues such as tank rupture or leakage resulting from overpressure, and core failures that disrupt the path of magnetic flux, are other reasons that can cause the GSU transformer to fail. The fault tree of a wind turbine GSU transformer, along with its associated basic events, is illustrated in Fig. 34, and the details of intermediate and basic events are provided in Tables 22 and 23.

Table 23
Basic failure events of a wind turbine transformer fault tree [71,77,78].

Basic events	Codes	λ (f/yr)
Tank rupture or leakage due to over pressure	e1	0.000333
Conductive part fails	e2	0.000700
Insulation part fails	e3	0.002592
Coil element fails	e4	0.000876
Magnetism lost	e5	0.000359
Instrumentation analog flux core fails	e6	0.002934
Paper element fails	e7	0.000876
Cooling System fails	e8	0.021286
Insulation contaminated	e9	0.000324
Cooling System fails	e10	0.007708
Insulation contaminated	e11	0.004380
Oil temperature indicator fails to initiate trip	e12	0.014629
Calibration oil level fails	e13	0.008059
Sensor temperature fails	e14	0.008059
Fails to initiate disconnect switch	e15	0.000876

7.2.4. Fault tree of wind turbine controls and sensors

Sensors are vital components within wind turbines, serving various purposes in monitoring and managing turbine performance. Among others, they include anemometers for wind speed measurement, wind direction sensors, temperature sensors for component monitoring, vibration sensors to detect mechanical issues, and tilt sensors to ensure safe angles. Load sensors help prevent overloading, while power and energy sensors track electrical output [80]. Other sensors control blade pitch, nacelle orientation, and lubricating oil quality. Environmental sensors provide operating conditions data, and communication sensors enable real-time monitoring and remote control. Here, communication sensors encompass various systems and components that facilitate data transfer and remote monitoring. This includes communication cables that transmit data between sensors and control systems, transmitters that convert sensor data into signals for transmission over networks, and communication modules that handle data exchange using technologies like Wi-Fi, cellular, or satellite. Additionally, data acquisition systems collect and manage data from various sensors, enabling its transmission to central control centers or remote monitoring locations. Together, these components ensure effective real-time monitoring and remote control of the wind turbine's operations. In cold climates, ice sensors detect ice accumulation, while lightning sensors protect the turbine from strikes. These sensors collectively optimize energy production, safety, and maintenance, offering valuable insights for turbine management [81–83]. Furthermore, the central component of a wind turbine that governs its performance is the control section. It is responsible for key functions, including aligning the rotor with the

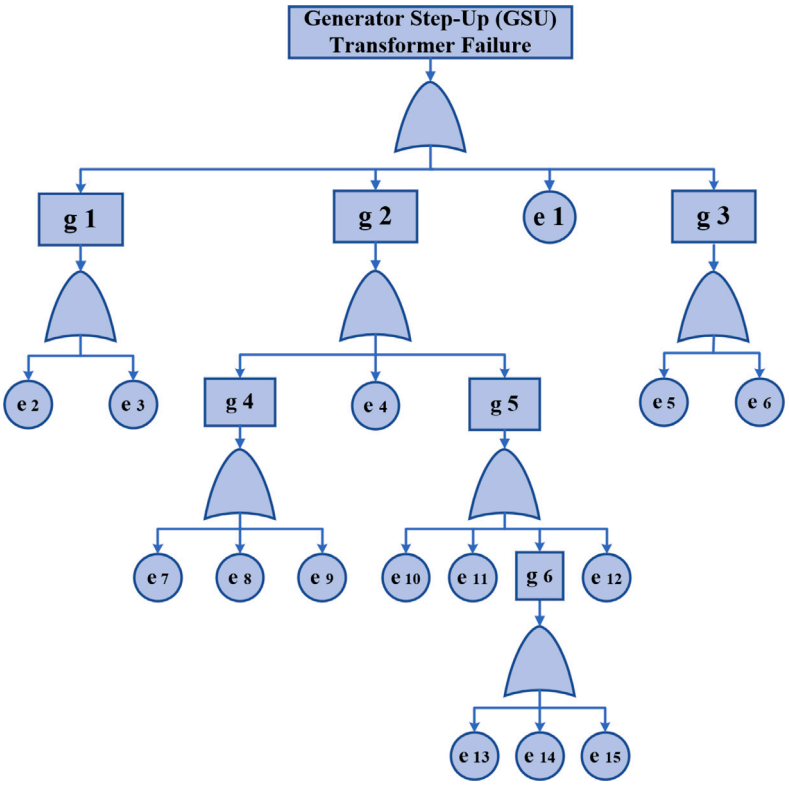


Fig. 34. Fault tree of a wind turbine transformer.

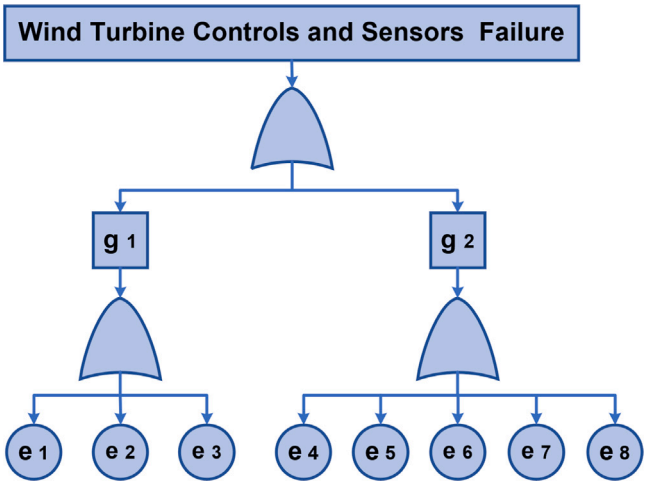


Fig. 35. Fault tree of a wind turbine controls and sensors.

Table 24
Intermediate events of a wind turbine controls and sensors.

Intermediate events	Code
Electrical failure	g1
Mechanical failure	g2

Table 25
Basic failure events of wind turbine controls and sensors.

Basic events	Code
Short circuit failure	e1
Open circuit failure	e2
Gate drive circuit	e3
Thermal related failure	e4
Humidity related failure	e5
Vibration related failure	e6
Dirt related failure	e7
Terminals damages	e8

wind, overseeing safety measures, integrating with the grid, and managing startup and shutdown procedures. Real-time data from sensors allows for fine-tuning to enhance efficiency. So, the control section functions as the turbine's control hub, optimizing energy capture and guaranteeing secure and dependable operation [84–86].

Failures in wind turbine controls and sensors can be attributed to issues stemming from both electrical and mechanical components. The primary electrical failures in wind turbine controls and sensors consist of short-circuit faults, open-circuit faults, and gate drive circuit faults. Meanwhile, the primary mechanical defect is corrosion resulting from exposure to salt mist and moisture. Ref. [14] provides detailed information on wind turbine controls and sensors failures, as presented in Tables 24 and 25, and their fault tree is depicted in Fig. 35.

7.2.5. Fault tree of a wind turbine gearbox

A wind turbine's gearbox plays a critical role in wind energy production by elevating the rotational speed of the rotor to the optimal rate for electricity generation through the generator. Gearboxes employ different gear types and ratios, which can include planetary, helical, and bevel gears, to effectively transfer the power [87]. Consistent maintenance is of utmost importance due to the considerable mechanical stresses and operational difficulties wind turbines encounter, given that gearbox malfunctions can result in costly repairs and downtime [25]. Moreover, certain turbine designs especially for offshore explore direct drive systems as gearbox substitutes. The wind energy industry maintains a strong focus on elevating gearbox performance to maximize wind turbine output. Within a gearbox, the most undesirable type of failures is a gear crack, as it frequently triggers more serious issues within the gear unit, ultimately causing the entire unit to break

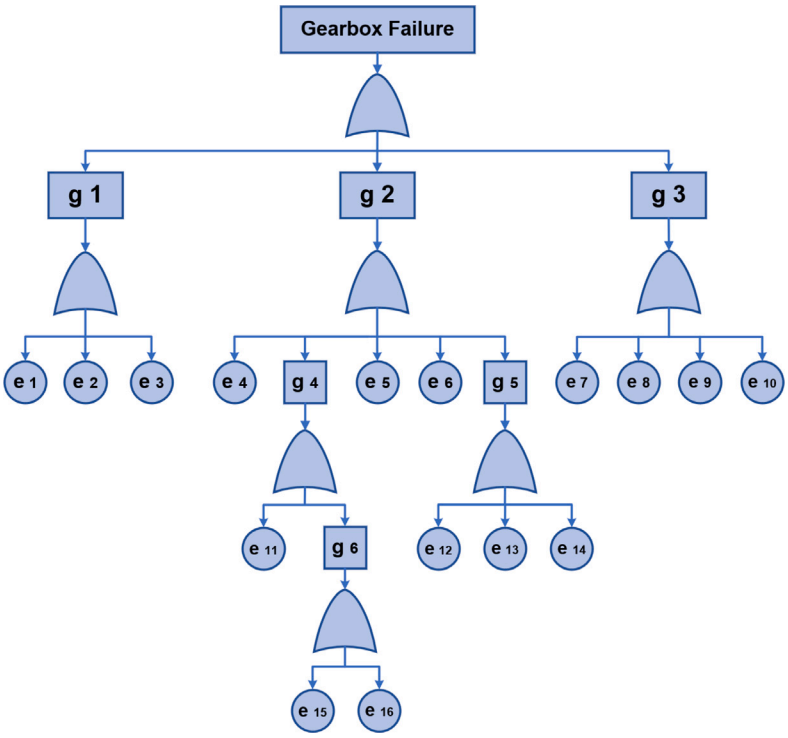


Fig. 36. Fault tree of a wind turbine gearbox.

Table 26
Intermediate events of a wind turbine gearbox fault tree.

Intermediate event	Codes
Lubrication exception	g1
Abnormal gear	g2
Bearing fault	g3
Tooth wear of gears	g4
Cracks in gears	g5
Offset of teeth gears	g6

Table 27
Basic failure events of a wind turbine gearbox fault tree [71,91,92].

Basic events	Codes	λ (f/yr)
Abnormal filter	e1	0.015768
Poor quality of lubrication oil	e2	0.015768
Dirt	e3	0.0126144
Abnormal vibration (Gearbox)	e4	0.0187464
Glued	e5	0.0021024
Pitting (gear)	e6	0.011388
Abnormal vibration (Gearbox)	e7	0.0187464
Corrosion of pins	e8	0.10512
Abrasive wear	e9	0.0876
Pitting (gear bearing)	e10	0.02628
Gear tooth deterioration	e11	0.002628
Excessive pressure	e12	0.00876
Excess temperature	e13	0.0021024
Fatigue	e14	0.002628
Poor design of teeth gears	e15	0.00876
Tooth surface defects	e16	0.002628

down [88]. The presence of salty air poses a threat to offshore wind turbine gearboxes, and one of the concerns is erosion. With the advancement of technology, direct-drive generators are expected to see broader adoption in offshore wind turbines as they offer the potential to reduce the life cycle cost (LCC) [89,90].

The causes of failures in wind turbine gearboxes often include issues associated with lubrication exceptions, abnormal gear conditions, bearing failures. Tooth wear on gears, cracks in gears, and offset teeth in gears are the factors that have key contributors to abnormal gear. The fault tree diagram for a wind turbine gearbox, encompassing both its basic and intermediate events, is visually represented in Fig. 36. Tables 26 and 27 provide a more in-depth understanding of these events based on [91,92].

7.2.6. Fault tree of a wind turbine yaw system

The wind turbine's yaw system holds a critical role in aligning the turbine rotor with the oncoming wind, ensuring the maximum capture of wind energy. This essential component typically comprises a yaw drive mechanism, which can be powered by either an electric motor or hydraulic system, along with a yaw controller that continuously monitors wind direction. If the controller detects a shift in the wind's direction, it triggers the yaw drive to adjust the rotor's orientation [93–95]. The effective operation of the yaw system is of paramount importance for optimizing energy production and safeguarding the turbine against excessive wind loads [96].

Wind turbine yaw system failures are primarily attributed to yaw motor failure, drive alarm, and meteorological unit fault, which are the most significant causal factors. The fault tree for a wind turbine yaw system failure is shown in Fig. 37. The intermediate events as well as basic events are listed in Tables 28 and 29, respectively.

7.2.7. Fault tree of a wind turbine blade system

Wind turbine blades serve as vital components in the utilization of renewable wind energy. Typically crafted from composite materials, such as a combination of fiberglass and carbon fiber, these blades are meticulously engineered to effectively capture wind energy [97]. Their form, size, and aerodynamic characteristics are finely tuned to optimize energy conversion while reducing resistance and turbulent effects. Contemporary wind turbine blades can vary in length, with certain examples extending beyond 80 m, and they are constructed

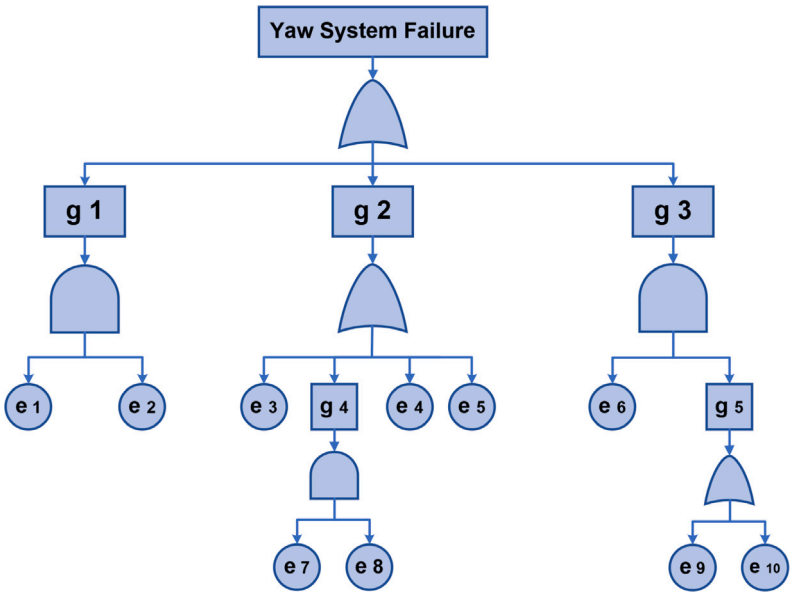


Fig. 37. Fault tree of a wind turbine yaw system.

Table 28
Intermediate events of a wind turbine yaw system fault tree.

Intermediate event	Codes
Yaw motor failure	g1
Drive alarm	g2
Meteorological unit fault	g3
Limit switch failure	g4
Meteorological sensors failure	g5

Table 29
Basic failure events of a wind turbine yaw system fault tree.

Basic events	Codes
Yaw motor fault	e1
Abnormal vibration	e2
Lightning module failure	e3
Yaw gear failure	e4
Cabinet switch trip	e5
Abnormal vibration	e6
Limit switch fault	e7
Limit slider fault	e8
Vane damage	e9
Anemometer damage	e10

to endure a range of environmental stress, including strong winds and turbulence, all while maintaining their structural integrity over many operational years [98]. Progress in both blade design and materials plays a crucial role in enhancing the efficiency and sustainability of wind energy production. Here, the blade system comprises the rotor hub, rotor bearings, and the blades themselves. The rotor hub, typically crafted from cast iron, functions as the component responsible for securing and stabilizing the blades during their rotation. However, the substantial loads it endures can result in issues such as clearance loosening and surface roughness. Surface roughness in a wind turbine hub can cause several issues leading to hub failure, including increased wear and friction on critical components, uneven load distribution, impaired lubrication, and heightened susceptibility to corrosion and erosion. These rough surfaces can also lead to imbalanced loading and excessive vibrations, which contribute to fatigue damage. Additionally, initial defects caused by roughness can propagate to other parts of the

Table 30
Intermediate events of a wind turbine blade fault tree.

Intermediate events	Codes
Structural fault	g1
Rotor system failure	g2
Tip damage	g3
Edges damage	g4
Shell damage	g5
Hub failure	g6
Bearings fault	g7
Imbalance of blade system	g8
Wear in bearings	g9

turbine, potentially causing more extensive failures. Proper manufacturing, maintenance, and inspection are crucial to mitigate these risks. The blades are connected to the rotor shaft through the hub and are supported by bearings within the rotor hub. The blade system failure originates from structural faults and rotor system failure. Structural issues in blades are primarily characterized by tip damages, edge damages, and shell damages, primarily stemming from factors such as cracks, erosion, delamination, debonding, as well as the strength and fatigue of the fibrous composite materials used in their construction. Furthermore, rotor bearings play a crucial role in enduring the dynamic forces and loads generated by the wind, but they are susceptible to damage from wear in the form of pitting, deformation of the outer surface and rolling elements, spalling, and overheating. A comprehensive fault tree illustrating the failure modes of the blades can be found in Fig. 38. Detailed information about intermediate events and basic events, along with their corresponding failure rates, is presented in Tables 30 and 31, respectively [28,65]. In Fig. 39, the depiction of a wind turbine blade system in a simplified model is presented.

7.2.8. Fault tree of a wind turbine pitch system

The wind turbine pitch system is one of the most vital components within modern wind turbines, playing a crucial role in ensuring both performance optimization and safety. The blade pitch system is directly linked to the turbine's blade assembly, where the physical structure of the blades is controlled by the pitch system to adjust their angle or "pitch". This connection between the two systems is key for adapting the turbine's operation to fluctuating wind conditions. The pitch system, whether it employs hydraulic or electric actuators,

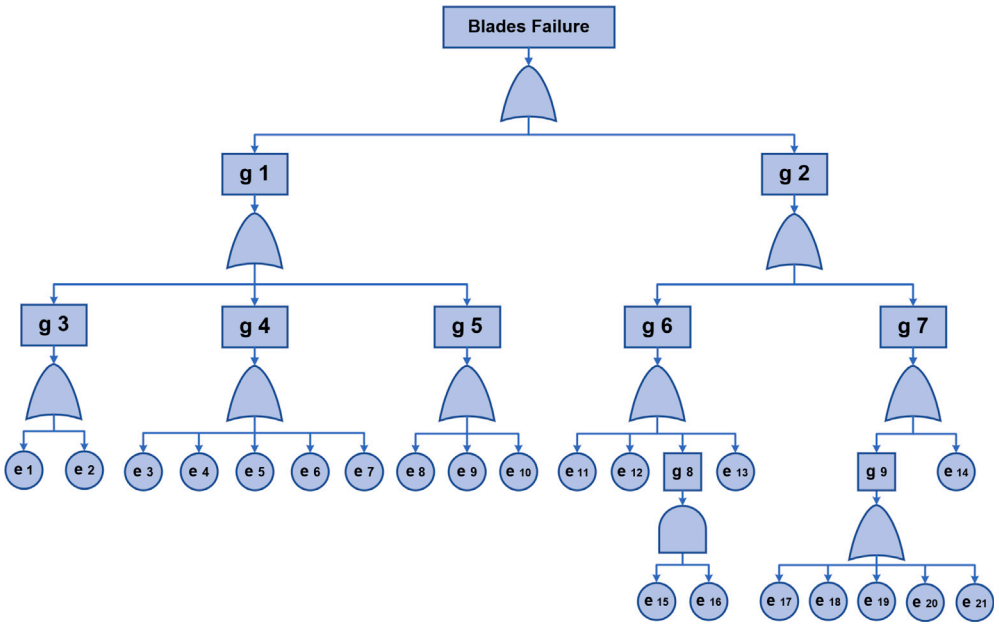


Fig. 38. Fault tree of a wind turbine rotor blades.

Table 31
Basic failure events of a wind turbine blades fault tree.

Basic events	Codes
Open tip	e1
Lightning strike on tip	e2
Cracks in the edge of blades	e3
Erosion in edges of blades	e4
Delamination in leading edges	e5
Delamination in trailing edges	e6
Debonding in edges of blades	e7
Delamination in shell	e8
Crack with structural damage	e9
Crack on the beam-shell joint	e10
Clearance loosening at root	e11
Cracks in the hub	e12
Surface roughness in the hub	e13
Cracks in bearings of rotor	e14
Mass imbalance in the hub	e15
Fault in pitch adjustment	e16
Corrosion of pins in bearings	e17
Abrasive wear in bearings	e18
Pitting in bearings of rotor	e19
Deformation	e20
Lubrication fault in bearings	e21

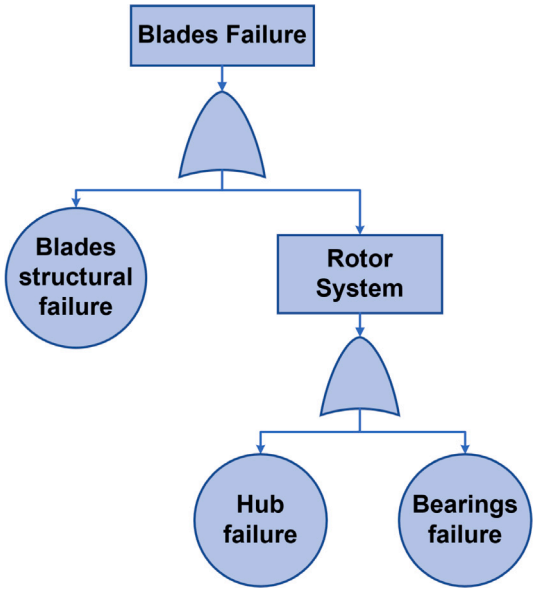


Fig. 39. Reduced fault tree of a wind turbine blades.

works in tandem with controllers to fine-tune the blade angles in real-time, maximizing power generation while simultaneously adapting to changing wind patterns. This not only enhances energy capture but also acts as a critical safety mechanism, preventing excessive rotor speeds during strong or gusty winds by feathering the blades to minimize stress on the turbine's structure [99,100].

In electric pitch systems, precise control of the blade angle is achieved through electric motors and sophisticated control algorithms that adjust each blade independently. Electric systems offer lower maintenance compared to hydraulic systems, as they do not rely on fluid-based mechanisms that are prone to leaks or pressure failures. Moreover, backup battery systems ensure continued functionality even during power outages, ensuring the turbine can safely shut down if needed. Hydraulic pitch systems, on the other hand, use pressurized fluids to enable the necessary blade adjustments. These systems, comprising pumps, valves, cylinders, and fluids, are also responsible for other critical functions in the turbine, such as yaw control, which allows the nacelle to align with the wind direction. By using pressurized fluid, hydraulic systems offer robust mechanical movement, ensuring precise control over the turbine's orientation and blade angle, which helps to optimize efficiency and increase the turbine's overall energy output [101].

Whether electric or hydraulic, the pitch system is essential for the reliable and efficient operation of a wind turbine, balancing performance and protection under diverse environmental conditions. In the context of pitch and hydraulic system failures, the hydraulic components emerge as the most susceptible elements within the entire assembly, with oil failure being the primary cause of malfunctions in the hydraulic system. The second most significant contributor to failures is the drive alarm, primarily stemming from issues like limit switch failures and lightning protection failures. Given its critical role

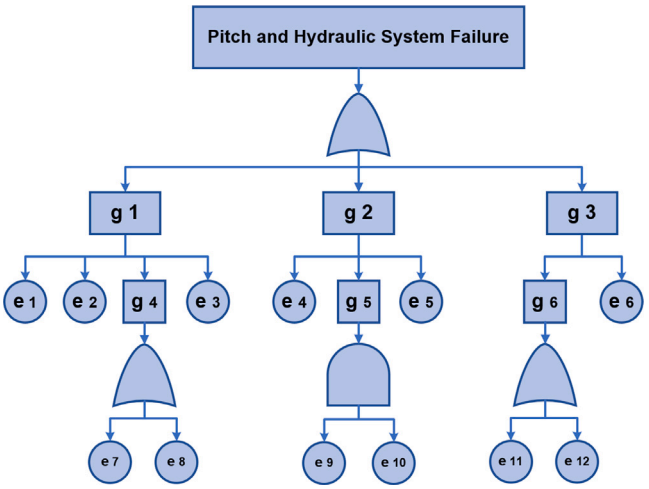


Fig. 40. Fault tree of a wind turbine pitch system.

Table 32
Intermediate events of a wind turbine pitch system fault tree.

Intermediate event	Codes
Hydraulic system failure	g1
Drive alarm	g2
Wrong blade angle	g3
Hydraulic oil failures	g4
Power failure	g5
Meteorological unit	g6

Table 33
Basic failure events of a wind turbine pitch system fault tree [71,103].

Basic events	Codes	λ (f/yr)
Hydraulic motor failure	e1	0.0876
Over pressure	e2	0.2628
Accumulator failure	e3	0.0596
Lighting protection failure	e4	0.0876
Limit switch fails	e5	0.0876
Abnormal vibration	e6	0.0188
Oil leakage	e7	0.4205
Filters failure	e8	0.0069
Power 1 failure	e9	0.4993
Power 2 failure	e10	0.4993
Vane damage	e11	0.0613
Anemometer damage	e12	0.1577

and elevated failure probability, the pitch and hydraulic system necessitate comprehensive reliability analysis and robust risk management measures to ensure optimal performance and minimize operational disruptions [102]. As provided in [103], Fig. 40 illustrates a fault tree representing the failure of the hydraulic and pitch systems. Furthermore, comprehensive details concerning intermediate events and basic events, inclusive of their associated failure rates, are available in Tables 32 and 33.

7.2.9. Fault tree of a wind turbine support system

Support structures for onshore and offshore wind turbines are meticulously designed to cater to the distinctive needs of their operating environments. Onshore wind turbines are typically affixed to towers, constructed from materials like steel or concrete, with the tower's height adjusted according to factors such as the turbine's rotor diameter and local wind conditions. Foundation choices for onshore turbines encompass shallow alternatives like mat or spread footings, as well as deeper options such as drilled piers, selected based on soil conditions and turbine size [104,105].

Table 34
Intermediate events of a wind turbine support system fault tree.

Intermediate events	Codes
Mooring system failure	g1
Tower failure	g2
Floating foundation failure	g3
Devices failure	g4
Extreme sea conditions	g5
Collapse due to environment	g6
Hit by dropped objects	g7
Watertight fault	g8
Other devices failure	g9
Pipe joint failure	g10
Fairlead failure	g11
Mooring lines broken	g12
Mooring lines breakage	g13
Mooring lines wear	g14
Accumulating wear	g15

On the other hand, offshore wind turbines face the difficulties posed by the harsh marine environment, requiring sturdy support structures to withstand these conditions. Shallow-water offshore wind turbines commonly rely on monopiles, massive steel columns driven into the seabed, while deeper waters demand jacket structures, steel frameworks with multiple legs anchored to the seabed. In exceedingly deep waters, floating platforms come into play, securely anchored yet capable of adapting to wave and current motion. Offshore wind farms additionally require intricate underwater cable systems to establish connections between turbines and offshore substations, which subsequently transmit generated electricity to the onshore grid. Offshore turbine maintenance poses heightened difficulties owing to their remote locations and constant exposure to marine conditions, calling for specialized vessels and equipment for servicing [106,107].

Fig. 41 provides a visual representation of a fault tree depicting the failure modes of the support structures. Based on this figure, mooring system failures are most commonly attributed to mooring line breakage. Anchor and fairlead failures follow as the second and third significant contributors to mooring system malfunctions. When examining tower failures, the highest probability is associated with collapse due to harsh environmental conditions. In the case of foundation failures, the most hazardous cause for floating foundation failure is being struck by objects dropped. In summary, the most prominent factor contributing to support structure failures is severe sea conditions, underscoring the importance of sophisticated weather forecasts and emergency response plans to mitigate losses. Detailed information regarding intermediate events and basic events, along with their corresponding failure rates, can be found in Tables 34 and 35 [17,108].

7.2.10. Fault tree of a HVAC transmission system

Fig. 42 depicts a simplified diagram of an offshore HVAC transmission arrangement [109]. In this configuration, the connection between the offshore and onshore substations is established using cross-linked polyethylene (XLPE) cables. Two offshore transformers increase the voltage from 33 or 66 kV at the collector end to transmission level options of 110, 150, or 220 kV. These parallel transformers are instrumental in boosting export power availability, rated at 60% of the nominal power of the Offshore Wind Power Plant (OWPP) [110]. However, implementing an HVAC transmission system for extended distances is technically challenging due to the heightened capacitance of the cables, leading to reactive current and substantial power loss. The amount of this reactive power is directly tied to the distance and the square of the system voltage [111,112].

The fault tree of the HVAC transmission system is depicted in Fig. 43. The HVAC transmission system faults originate from AC bus failure, large transformer failure, and HVAC cable or underwater cable failure. The AC bus failure comes from the bus itself, large switch

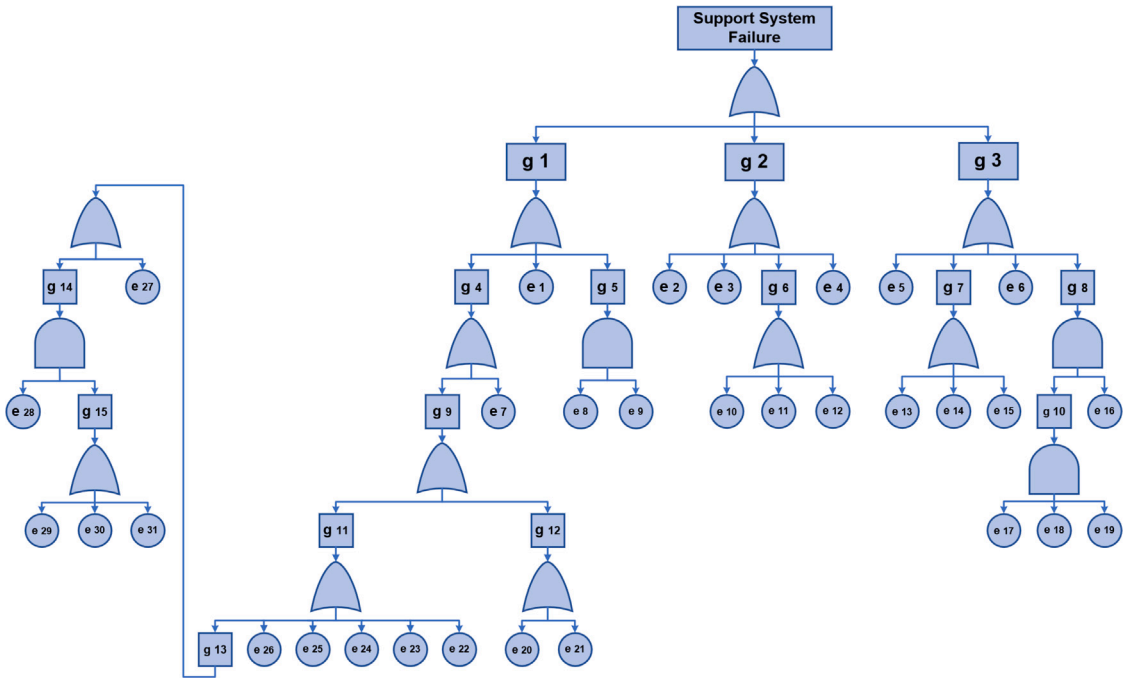


Fig. 41. Fault tree of a wind turbine support system.

Table 35
Basic failure events of a wind turbine support system fault tree [71,103].

Basic events	Codes	λ (f/yr)
Human error	e1	0.05256
Resonance	e2	0.04380
Faulty welding of tower	e3	0.06132
Material fatigue	e4	0.09636
Pillar damage	e5	0.04380
Capsize	e6	0.00876
Anchor failure	e7	0.15768
Poor operation environment	e8	0.68328
Insufficient emergency measures	e9	0.00876
Strong wind and/or wave	e10	0.43800
Lightning strike	e11	0.06132
Storm	e12	0.48180
Typhoon	e13	0.87600
Plane crash	e14	0.00876
Bird collision	e15	0.04380
Insufficient detection	e16	0.07577
Pipe joint corrosion	e17	0.11388
Pipe joint weld defect	e18	0.02628
Pipe joint fatigue	e19	0.02628
Fairlead corrosion	e20	0.08760
Fairlead fatigue	e21	0.14892
Transitional chain wear	e22	0.08848
Friction chain wear	e23	0.06071
Mooring winch failure	e24	0.07008
Buoys friction chain wear	e25	0.03671
Anchor pickup device damaged	e26	0.04871
Abnormal stress	e27	0.35653
Invalid maintenance	e28	0.33112
Mooring lines wear	e29	0.14016
Mooring lines fatigue	e30	0.14892
Mooring lines corrosion	e31	0.04712

failure, and feeder system failure. The fault tree of feeder failure is shown in Fig. 44. In this diagram, $WT_1, WT_2, \ldots,$ and WT_n represent wind turbine failures. Tables 36 and 37 demonstrate the basic and intermediate events of the HVAC transmission system fault tree and

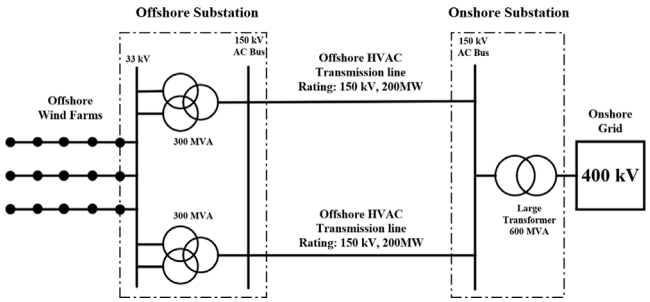


Fig. 42. Fundamental setup of an HVAC solution.

Table 36
Intermediate events of HVAC transmission system fault tree.

Intermediate events	Codes
AC bus failure	g1
Feeder system failure	g2

Table 37
Basic failure events of HVAC transmission system fault tree [71,113].

Basic events	Codes	λ (f/yr)	μ (r/yr)
HVAC transmission line failure	e1	0.014100	3.04
Large transformer failure	e2	0.003200	0.454
AC bus failure	e3	0.000125	0.0084
Large switch failure	e4	0.009600	8.75

the failure rates as well as repair time of HVAC transmission system different parts.

7.2.11. Fault tree of a HVDC transmission system

HVDC transmission stands out as the favored choice for modern long-distance offshore wind power transmission. Its primary advantage lies in the low transmission loss, with approximately 3% loss per 1,000 km in transmission power and about 1.5% loss in the power of the converter station concerning the nominal transmission power [114]. The

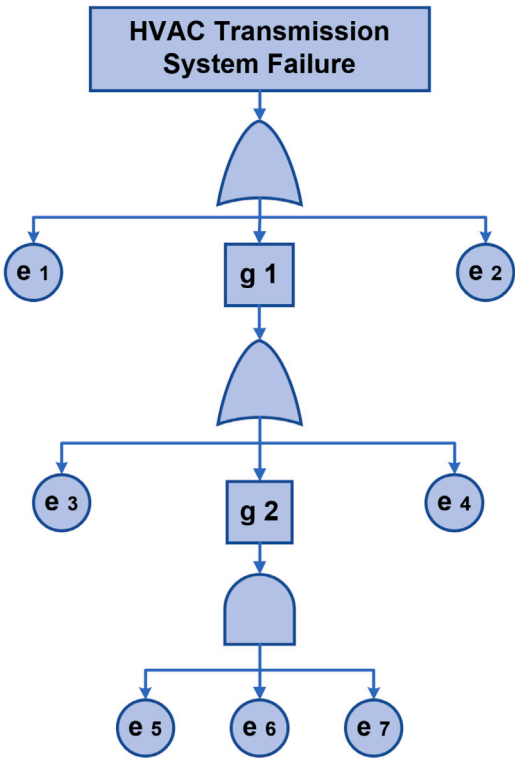


Fig. 43. Fault tree of a HVAC transmission system.

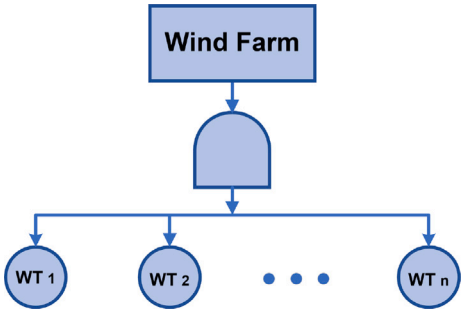


Fig. 44. Fault tree of a feeder.

straightforward layout of an HVDC transmission topology is illustrated in Fig. 45 [109].

Several topologies are employed for the HVDC transmission of offshore wind power. These include thyristor-based HVDC and Voltage Source Converter-HVDC (VSC-HVDC) technologies. Although thyristor-based HVDC is not interesting for wind farms nowadays. Each approach presents distinct features and advantages, contributing to the versatility and efficiency of offshore wind power transmission systems [115]. The voltage source converter-HVDC (VSC-HVDC) exhibits a notable advantage over traditional thyristor-based systems. This technology represents a significant advancement in HVDC systems, providing enhanced control capabilities and reliability [116].

The fault tree of VSC-HVDC transmission system is depicted in Fig. 46. The VSC-HVDC transmission system faults are originated from DC bus failure, large transformer failure, HVDC transmission line failure and DC/AC converter failure. The DC bus failure comes from AC bus failure, large switch failure, AC/DC converter and feeder system failure. The fault tree of feeder failure is shown in Fig. 44. Tables 38 and 39 demonstrate the basic and intermediate events of VSC-HVDC transmission system fault tree and the failure rates as well as repair time of HVDC transmission system different parts.

Table 38
Intermediate events of SVC-HVDC transmission system.

Intermediate events	Codes
DC bus failure	g1
Feeder system failure	g2

Table 39
Basic failure events of VSC-HVDC transmission system [71,113].

Basic events	Codes	λ (f/yr)	μ (r/yr)
HVDC transmission line failure	e1	0.012300	3.0400
Large transformer failure	e2	0.003200	0.0454
DC/AC converter failure	e3	0.029800	3.2330
AC bus failure	e4	0.000125	0.0084
Large switch failure	e5	0.009600	8.7500
AC/DC converter failure	e6	0.029800	3.2330

Table 40
Intermediate events of a wind farm transformer fault tree.

Intermediate event	Codes
On load tap changer failure	g1
Bushing failure	g2
Winding fails to induce electricity	g3
Core fails to be path of magnetic flux	g4
Drive mechanism failure	g5
Paper insulation fails	g6
Oil insulation fails	g7
Oil level gauge fails to initiate trip	g8

7.2.12. Fault tree of a wind farm transformer

The wind farm transformers act as a vital link between wind turbines and the distribution grid [117]. The main purpose of the wind farm transformer is to convert the variable and sometimes unpredictable electrical output produced by numerous wind turbines into a steady and controllable format that can be seamlessly integrated into the larger power grid. This process encompasses adjusting the generated electricity to meet the necessary voltage levels for effective transmission and distribution. Moreover, wind farm transformers are essential for identifying and isolating potential electrical issues within the wind farm, ensuring a dependable electricity flow to the grid. This, in turn, enhances the overall stability and efficiency of the renewable energy generation system [118].

Wind farm transformer faults can occur due to a range of factors, including on-load tap changer failure, bushing failure, winding failure, tank failure, and core failure [79]. On-load tap changer failure occurs because of tap selection switch failure, control device failure, and drive mechanism failure that comes from power supply failure, motor failure, and driving parts failure. The rest of the failures are similar to wind turbine GSU transformer failures mentioned in Section 7.2.3. The fault tree of a wind farm transformer, along with its associated basic events, is illustrated in Fig. 47, and the details of intermediate and basic events are provided in Tables 40 and 41. Notice, information regarding the average time required to repair different components of a wind farm transformer is insufficient.

7.2.13. Fault tree of a circuit breaker

The high voltage circuit breaker holds significant importance as a crucial component within power system switchgear. Its pivotal role in regulating and safeguarding power systems underscores its critical contribution to ensuring the stable functioning of these systems [119]. Circuit breaker failures can result in power supply interruptions and significant financial losses. The key to ensuring the safe and reliable operation of the power system lies in conducting a precise reliability assessment of the circuit breaker. This assessment serves as a crucial prerequisite for upholding the security and dependability of the overall system [120,121].

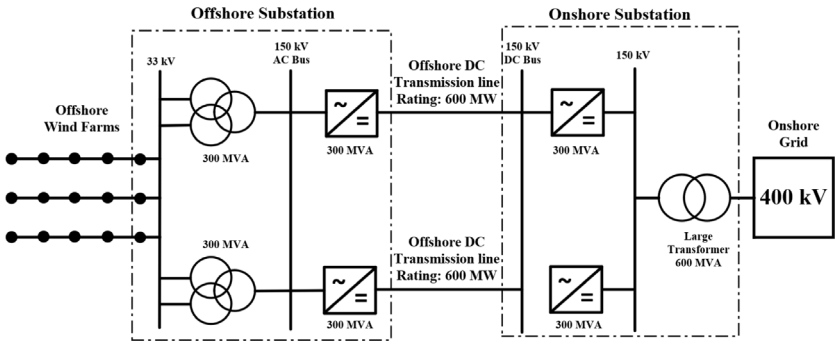


Fig. 45. Fundamental setup of a HVDC solution.

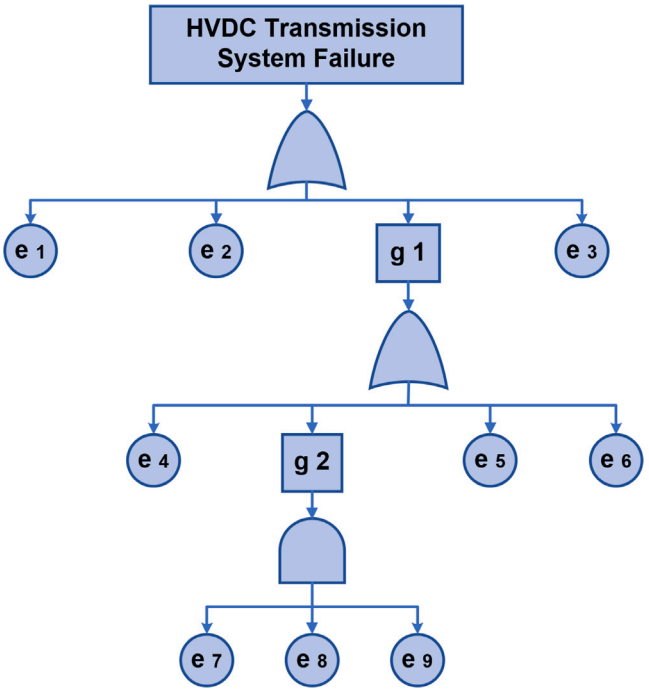


Fig. 46. Fault tree of a VSC-HVDC transmission system.

Table 41
Basic failure events of a wind farm transformer fault tree [71,79].

Basic events	Codes	λ (f/yr)
Tank rupture or leakage due to over pressure	e1	0.000333
Control device fails	e3	0.008760
Conductive part fails	e4	0.070080
Insulation part fails	e5	0.002592
Coil element fails	e6	0.000876
Magnetism lost	e7	0.000359
Instrumentation analog flux core fails	e8	0.002934
Power supply fails	e9	0.026280
Motor fails	e10	0.026280
Driving parts fails	e11	0.028032
Paper element fails	e12	0.000876
Cooling System fails	e13	0.021286
Insulation contaminated	e14	0.000324
Cooling System fails	e15	0.007708
Insulation contaminated	e16	0.004380
Oil temperature indicator fails to initiate trip	e17	0.014629
Calibration oil level fails	e18	0.008059
Sensor temperature fails	e19	0.008059
Fails to initiate disconnect switch	e20	0.000876

Table 42
Intermediate events of circuit breaker.

Intermediate events	Codes
Mechanical Failure	g1
Electrical Failure	g2

Table 43
Basic failure events of a SF6 circuit breaker fault tree [71].

Basic events	Codes	λ (f/yr)
Contact welding failure	e1	0.001
Spring break failure	e2	0.0005
Arcing damage	e3	0.002
Overload failure	e4	0.005

The fault tree of a circuit breaker is shown in Fig. 48. The fault tree for a circuit breaker failure includes two main branches: Mechanical failure and Electrical failure. Mechanical failure can result from contact welding, where contacts become stuck, or spring breakage, preventing proper operation. Electrical failure includes arcing damage from operation and overload due to excessive current. These basic events collectively contribute to the top-level event of circuit breaker failure [122,123]. Tables 42 and 43 demonstrate the intermediate and basic events of a circuit breaker fault tree, respectively.

8. Case study

In this review examining the dependability of a wind farm, an analysis using fault tree methodology in earlier sections is performed to evaluate different elements, such as wind turbines, transformers, HVDC/HVAC systems, and circuit breakers. It is important to highlight that a Weibull distribution function is utilized to model the wear-out failure rate. The hazard function, denoted as $h(t)$, is outlined as follows:

$$h(t) = \lambda_{w-x}(t) = \beta\alpha^{-\beta}t^{\beta-1} \tag{13}$$

In the context of the Weibull distribution employed for modeling the wear-out failure rate of component x, the symbols λ_{w-x}, α, and β represent the wear-out failure rate of part x, as well as the scale and shape parameters of the Weibull distribution, respectively [124].

Although certain components have available data on failure rates, mean times to repair, scale parameters, and shape parameters, there is a lack of information for other components. To tackle this issue, assumed or standard data are utilized for components that lack specific information. The presentation of this data is detailed in [125]. Hence, the approach to compute the overall failure rate and repair rate entails amalgamating data from components with known information and those with assumed data. Furthermore, it is assumed that the various parts of the wind farm have a lifespan of 30 years. Formulas ((6), (7),

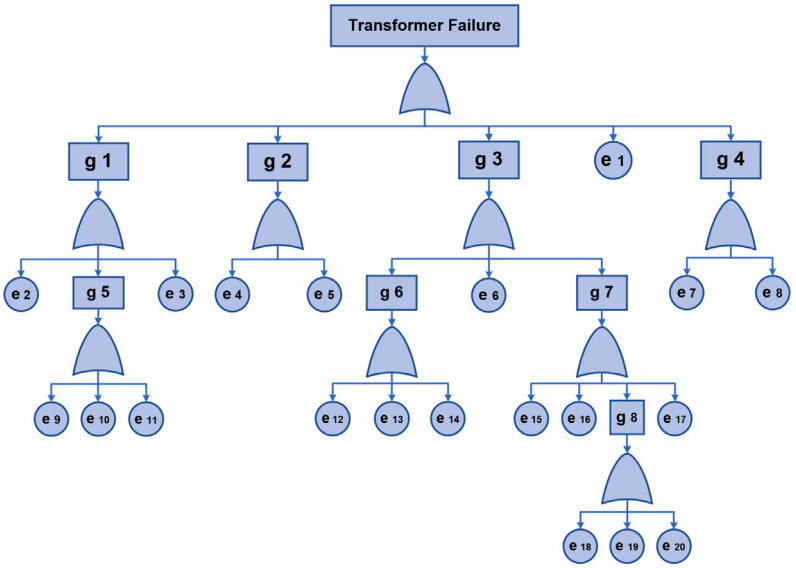


Fig. 47. Fault tree of a wind farm transformer.

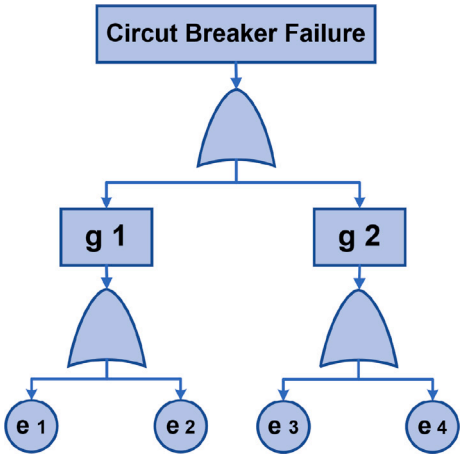


Fig. 48. Fault tree of a SF6 circuit breaker.

Table 44
Failure rate ($\lambda(f/yr)$) and repair rate ($\mu(r/yr)$) of a wind farm.

Year	λ (f/yr)	μ (r/yr)	Year	λ (f/yr)	μ (r/yr)
1	5.4077	5.1987	16	7.1621	5.2048
2	5.4212	5.2070	17	7.4321	5.1986
3	5.4453	5.2168	18	7.7259	5.1922
4	5.4812	5.2254	19	8.0442	5.1856
5	5.5299	5.2318	20	8.3879	5.1787
6	5.5922	5.2357	21	8.7578	5.1715
7	5.6692	5.2376	22	9.1546	5.1642
8	5.7617	5.2376	23	9.5792	5.1565
9	5.8706	5.2362	24	10.0322	5.1486
10	5.9967	5.2337	25	10.5146	5.1404
11	6.1409	5.2303	26	11.0272	5.1319
12	6.3040	5.2261	27	11.5705	5.1230
13	6.4869	5.2214	28	12.1456	5.1138
14	6.6903	5.2162	29	12.7531	5.1042
15	6.9152	5.2106	30	13.3939	5.0942

(10), (11) and (13)) are utilized to calculate the wind farm's overall reliability metrics. The outcomes, illustrated in Table 44, reveal the comprehensive reliability of a wind farm by presenting the overall failure rate and repair rate for each year.

9. Conclusion

In conclusion, this review has systematically examined and synthesized extensive data on the reliability of crucial components in renewable energy systems based on different surveys. The analysis covered wind turbines, transformers, HVDC/HVAC systems, and circuit breakers, providing a comprehensive global perspective on failure rates, downtimes, and critical components. Although there is a lack of comprehensive data on available surveys, the findings offer valuable insights into the performance of individual wind turbine components, transformer failures in diverse global regions, cable systems across varying environments and voltage levels, and circuit breaker reliability across multiple countries. The incorporation of international surveys, industry databases, and a fault tree analysis of wind farms enhances the robustness of the study. This comprehensive examination contributes to the understanding of reliability challenges and provides a foundation for informed decision-making in the operation, maintenance, and planning of wind farms.

For future research, it would be beneficial to explore advanced data analysis methods, such as machine learning and predictive analytics, to uncover deeper insights from reliability data. Addressing unresolved issues like common-cause failures and component interdependencies is also crucial. Additionally, developing more detailed fault tree models and investigating the impact of emerging technologies, such as improved sensors and predictive maintenance, could provide valuable insights.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Seyed Amir Hosseini reports financial support was provided by TotalEnergies OneTech. Saeed Peyghami reports financial support was provided by TotalEnergies OneTech. Frede Blaabjerg reports financial support was provided by TotalEnergies OneTech. Arsim Ahmedi reports financial support was provided by TotalEnergies OneTech. If there

are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.rser.2024.115227.

Data availability

I put the references for the data.

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